



Archeology of interstellar archipelagos

Stitching together the interstellar medium in and
around molecular clouds

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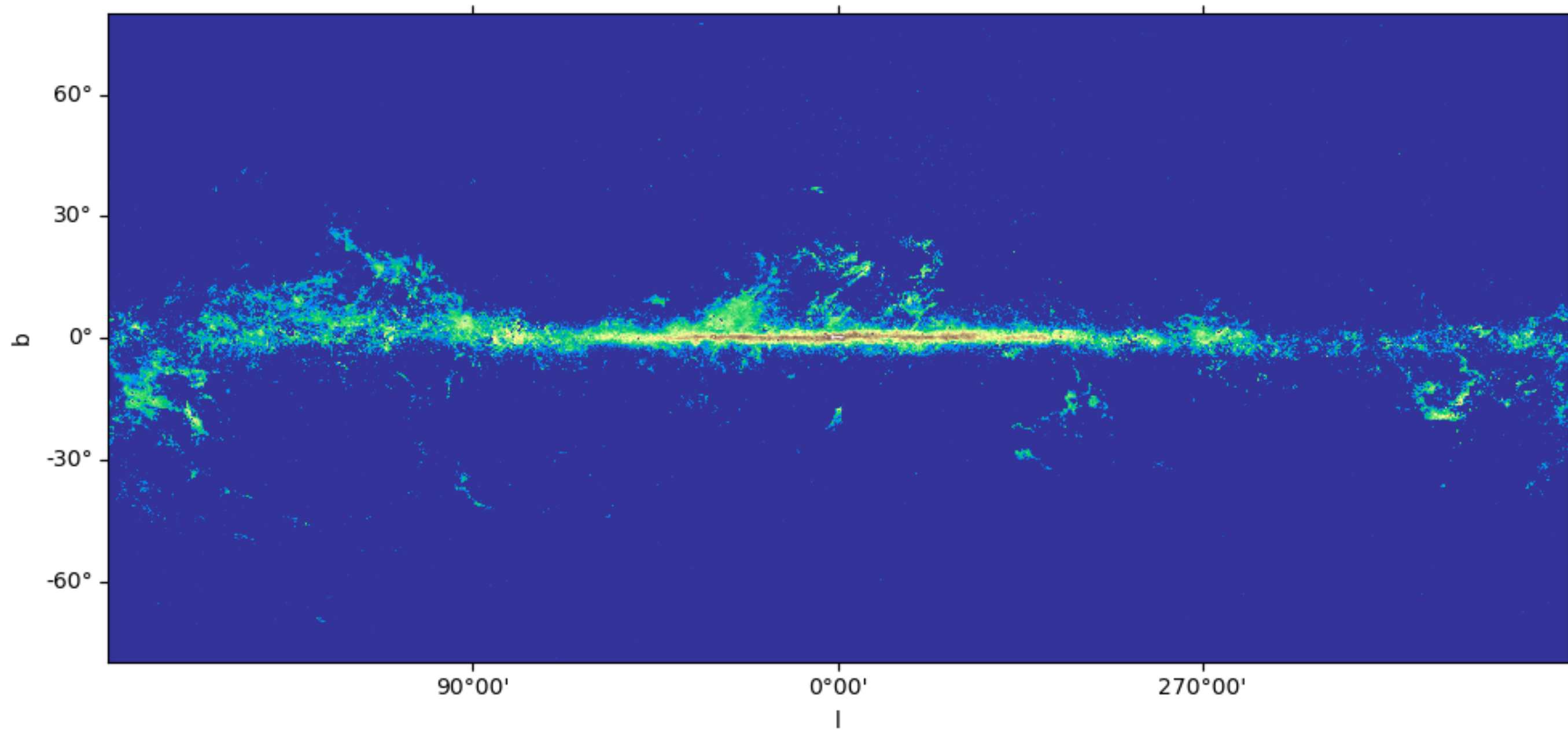
Roma, Italia

In collaboration with:

C. Zucker, J.E.G. Peek, E. Zari, P. Goldsmith, M. Heyer
ECOGAL: S. Molinari, R.S. Klessen, P. Hennebelle, L. Testi,
S.C.O. Glover, A. Traficante, E. Schisano, D. Elia,
P. Girichidis, M. Sormani, R., Treß, T. Colman, R. J. Smith

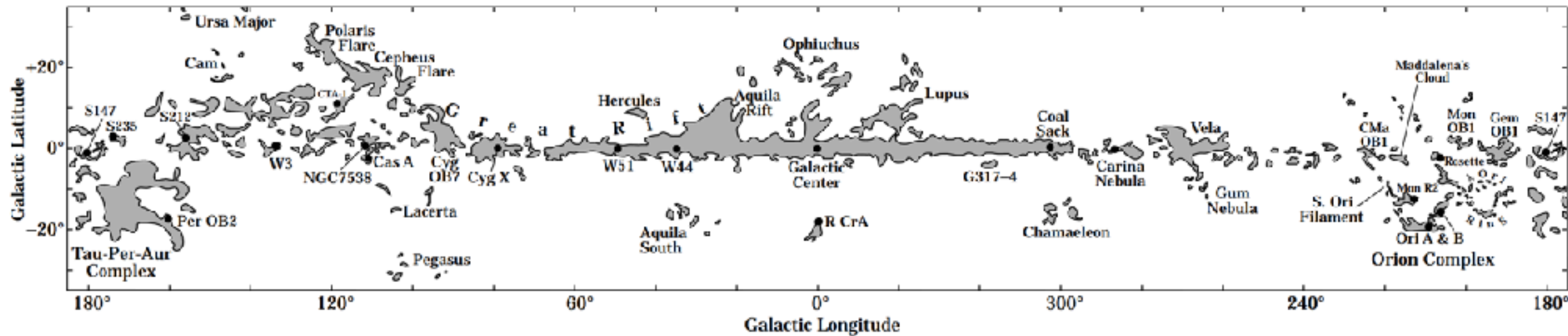
Carbon monoxide (CO) emission

Planck 2013 results. XIII. A&A (2014)



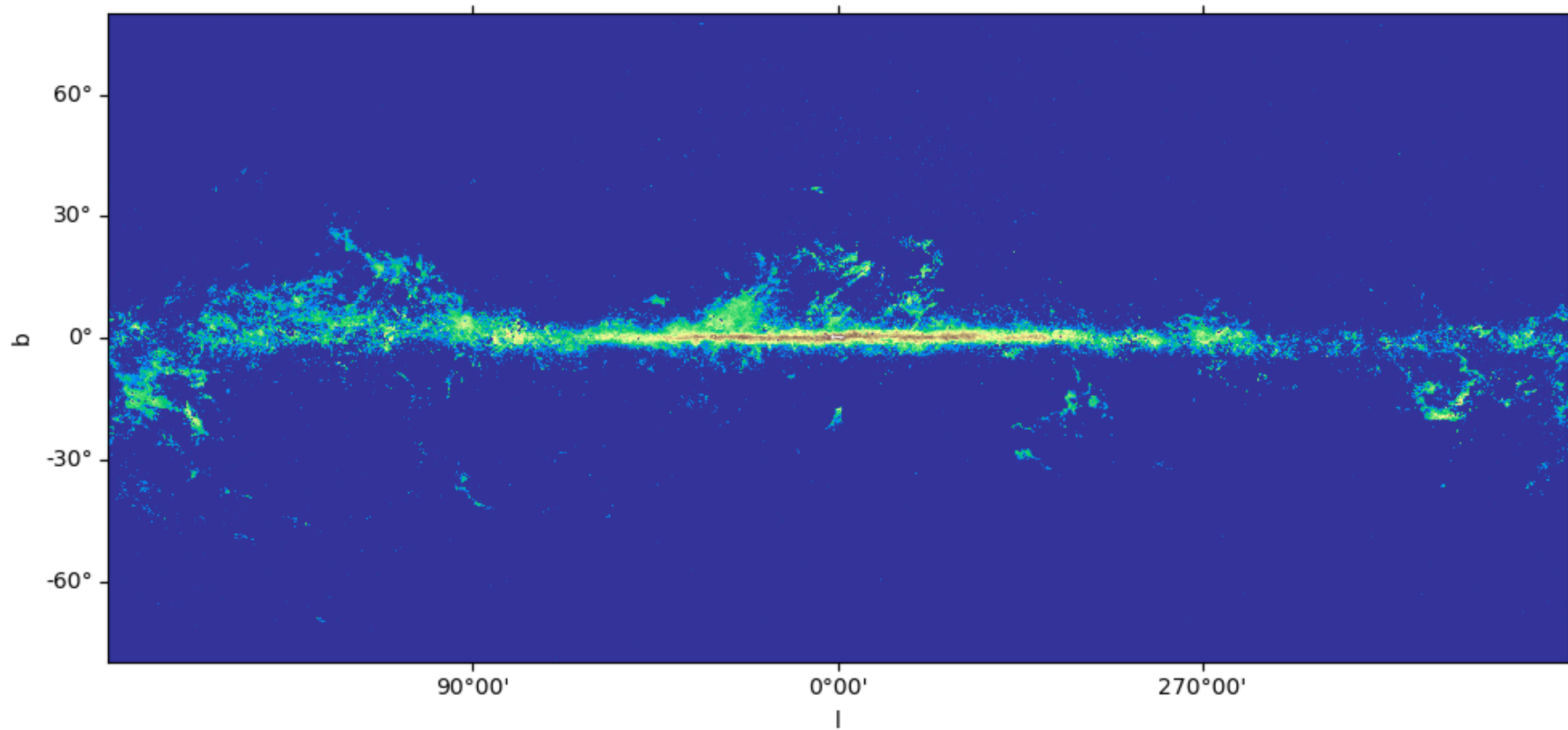
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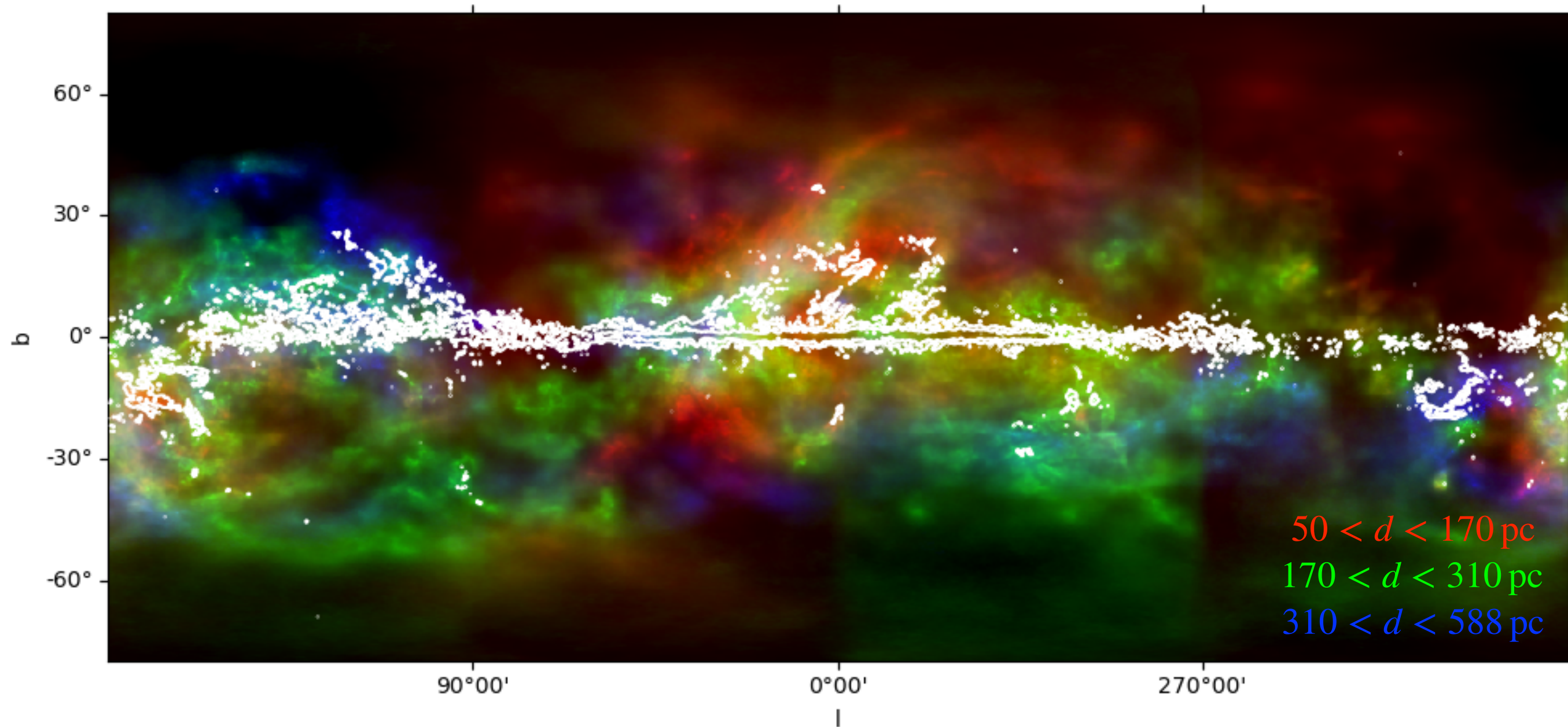
Carbon monoxide (CO) emission

Planck 2013 results. XIII. A&A (2014)



Local 3D dust density reconstruction

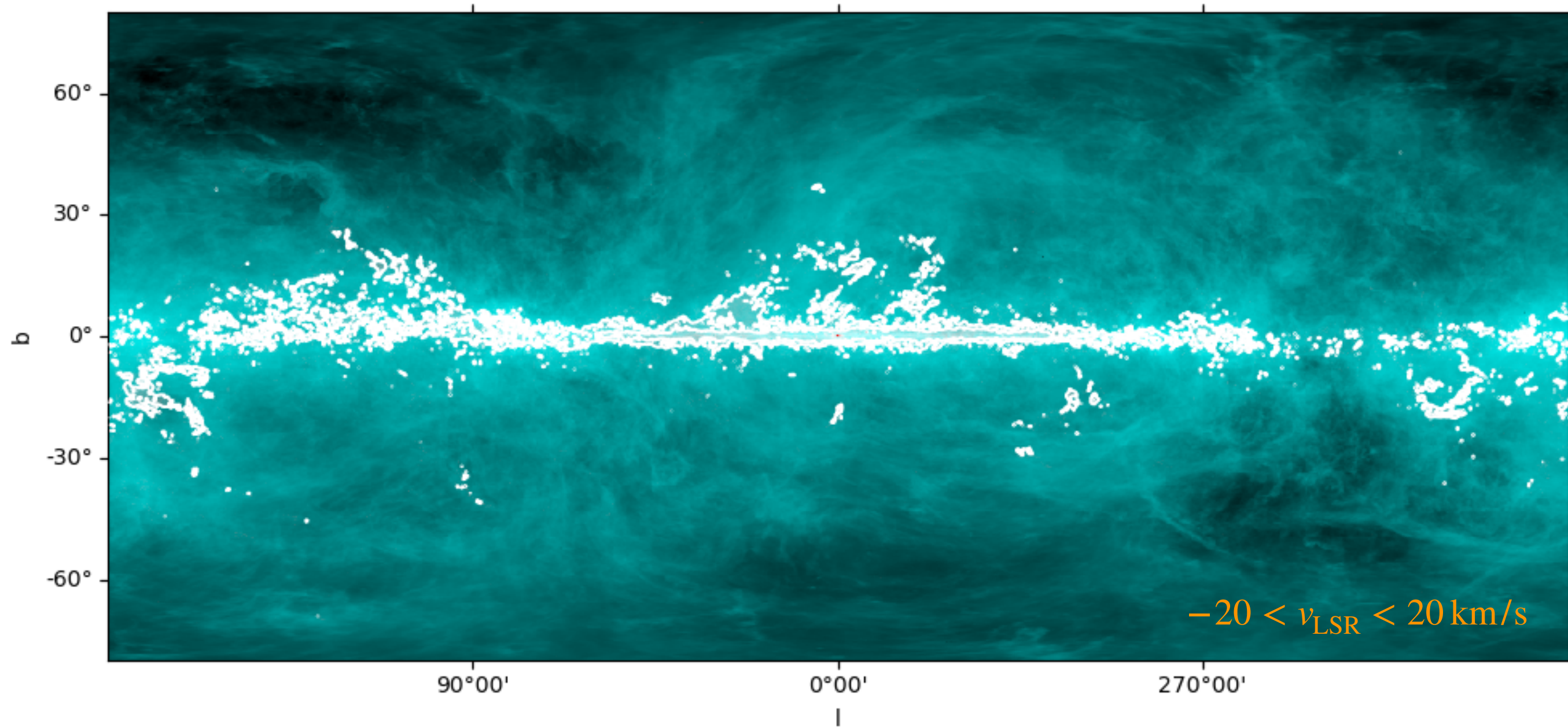
Leike et al. A&A (2020); Leike et al. A&A (2019);



Neutral atomic hydrogen (HI) emission

HI4PI Collaboration. A&A (2016)

McClure-Griffiths, Stanimirović, & Rybarczyk. ARA&A (2023)

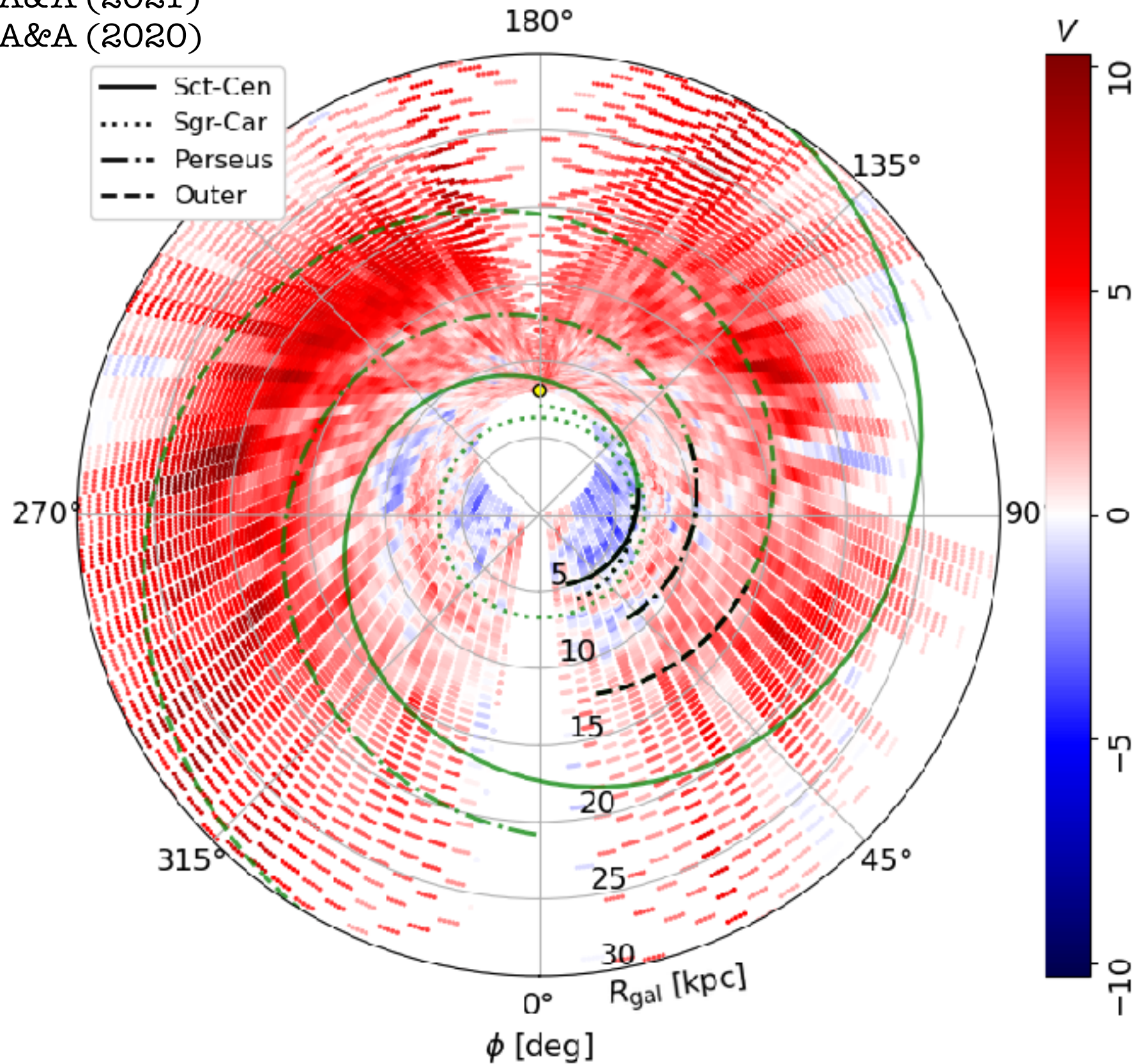


HI filament orientation trilogy

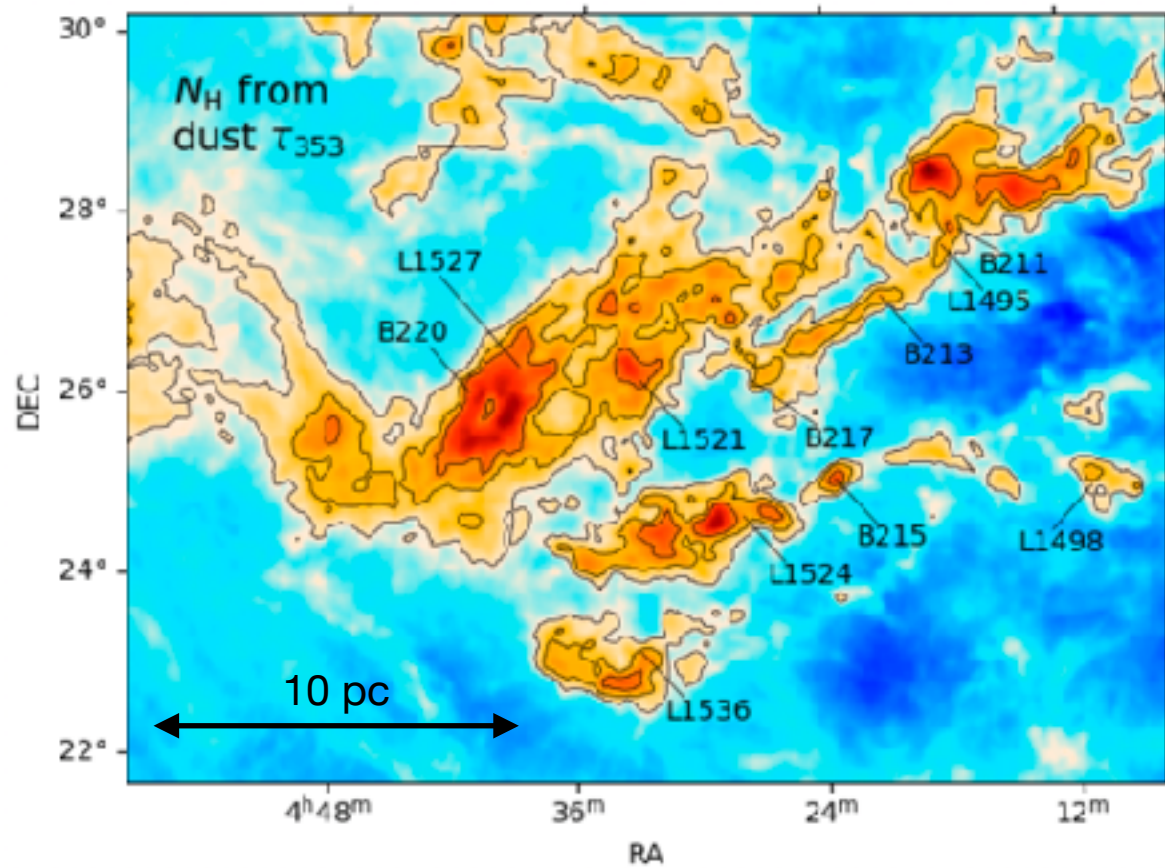
Soler, J.D., MAMD, et al. A&A (2022)

Soler, J.D., et al. A&A (2021)

Soler, J.D., et al. A&A (2020)



Two examples of interstellar archeology

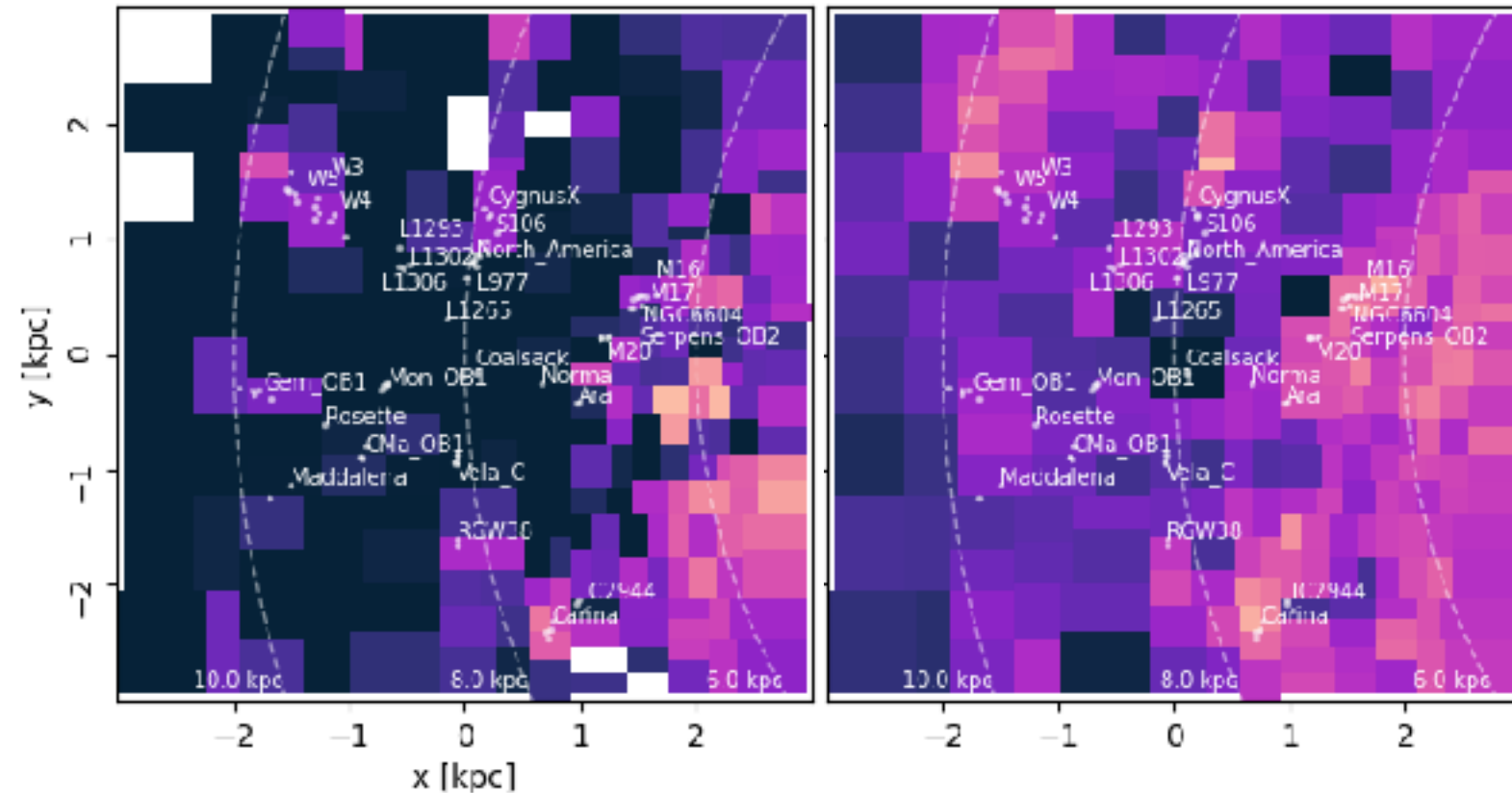


Dynamics of the Taurus molecular cloud

Soler, Zucker, Peek &
the ECOGAL collaboration
A&A (2023)

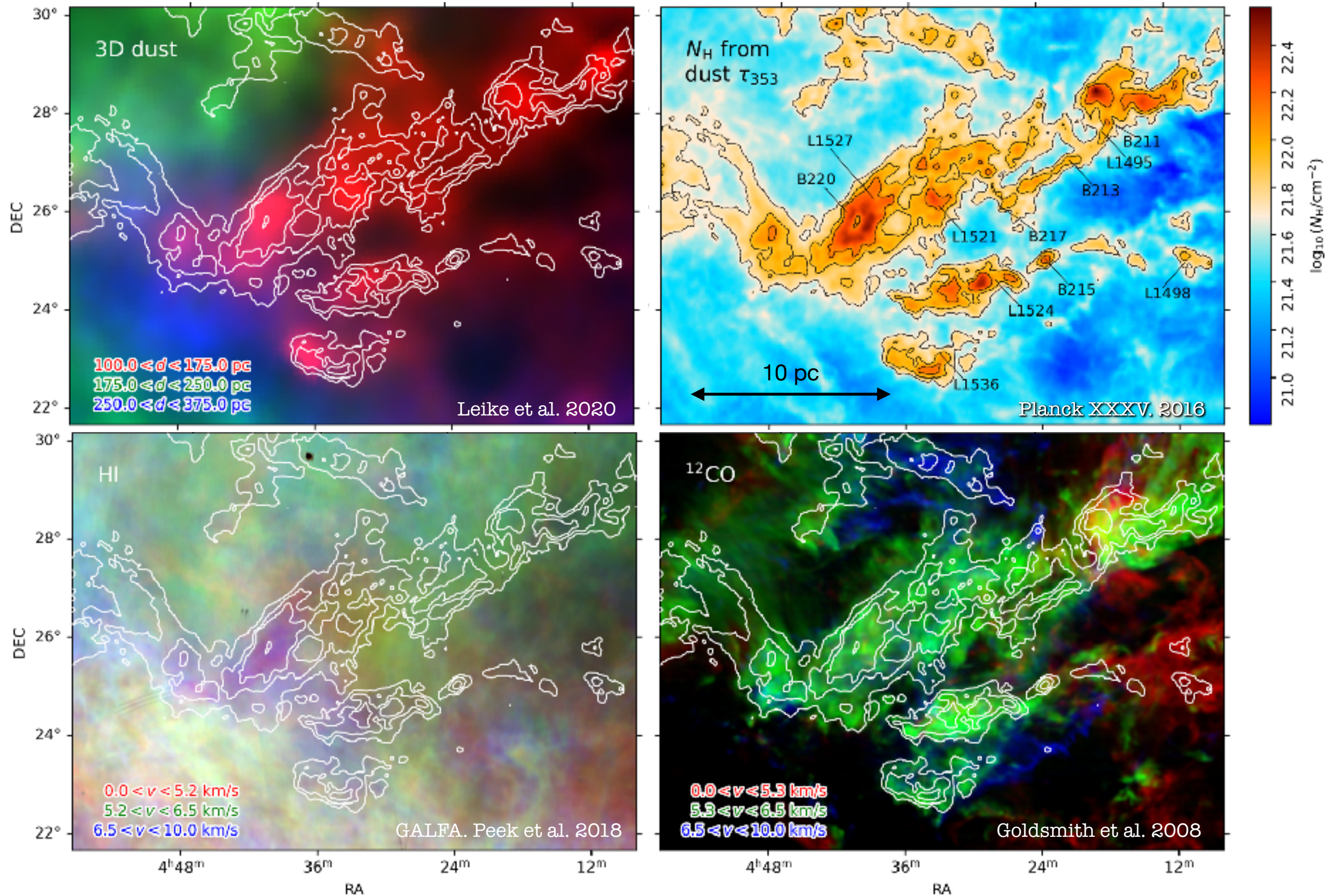
Star formation from dust thermal emission and high-mass stars

Soler, Zari, Elia &
the ECOGAL collaboration
About to be submitted.



Dynamics of the Taurus molecular cloud

Soler, Zucker, et al. A&A (2023)



Histogram of oriented gradient (HOG) method

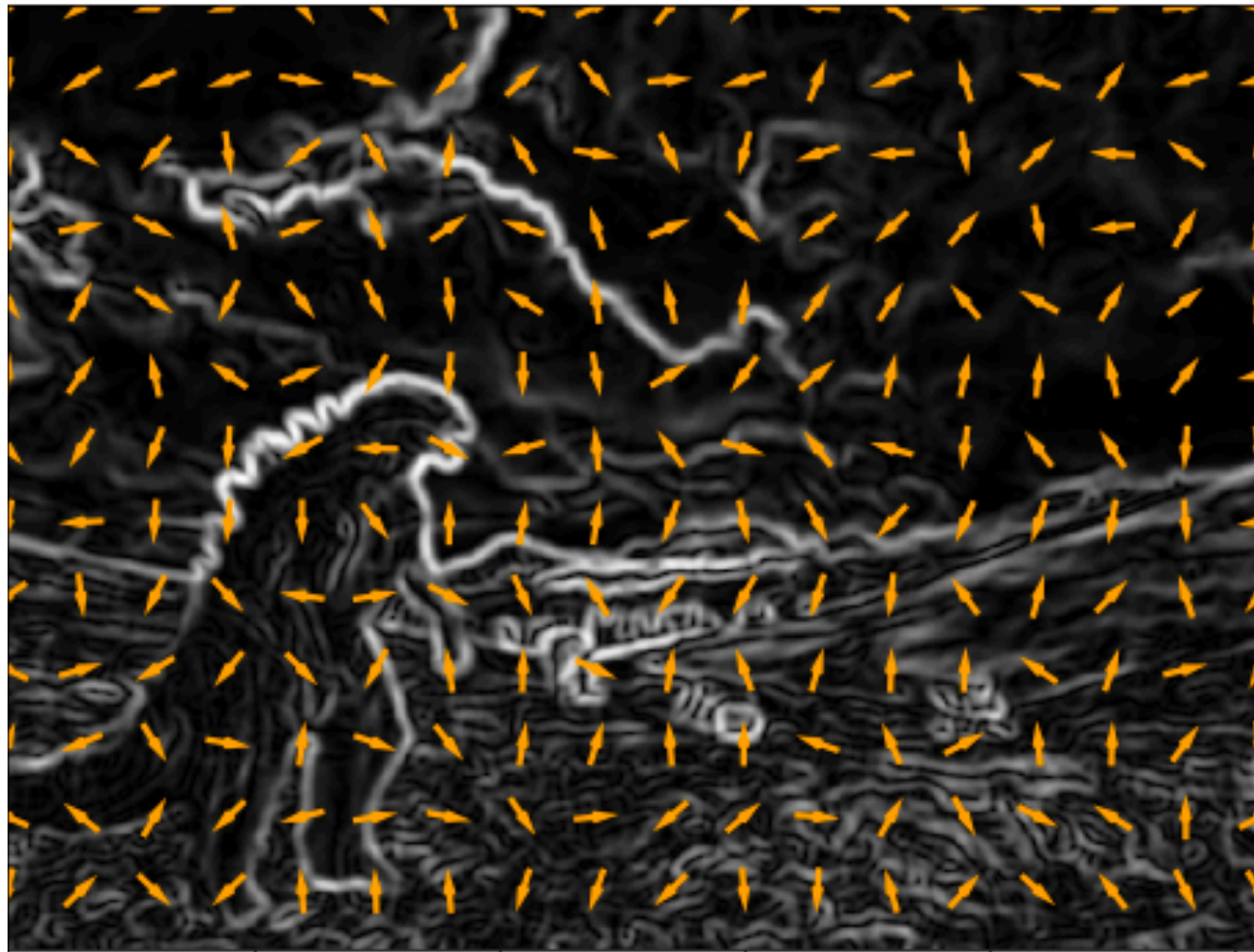
Soler, Beuther, et al. A&A (2019)



<https://github.com/solerjuan/astrohog>

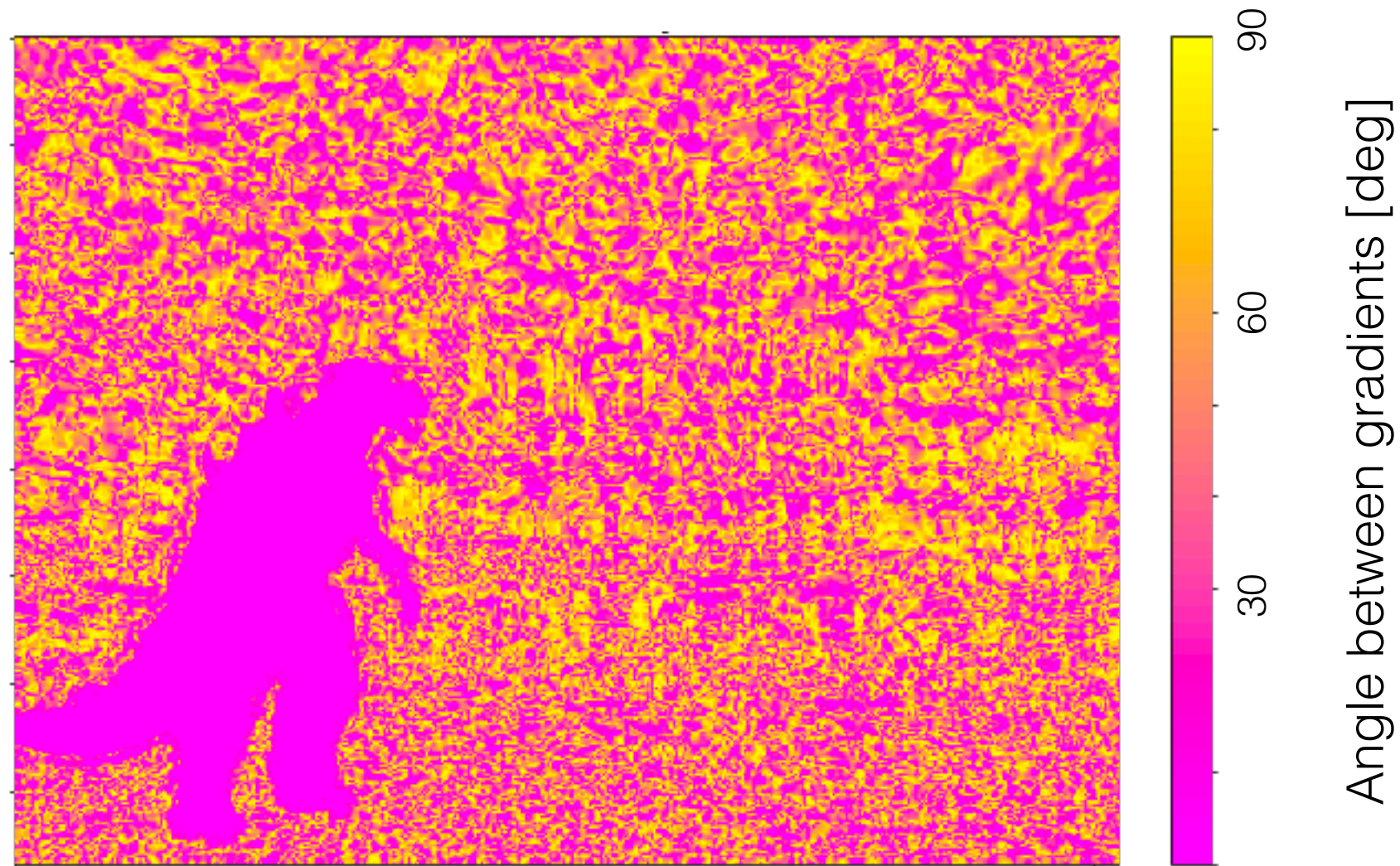
Histogram of oriented gradient (HOG) method

Soler, Beuther, et al. A&A (2019)



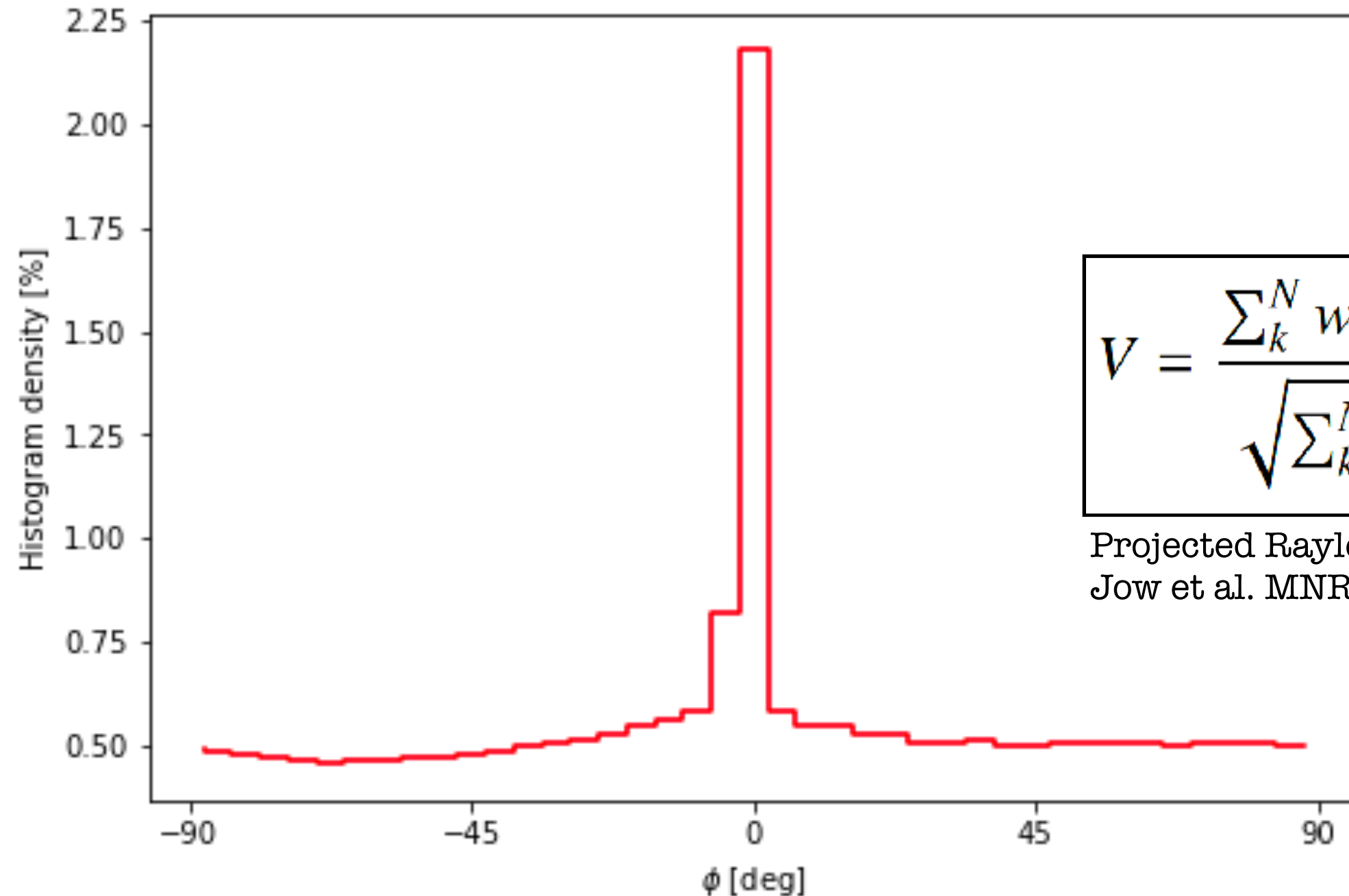
Histogram of oriented gradient (HOG) method

Soler, Beuther, et al. A&A (2019)



Histogram of oriented gradient (HOG) method

Soler, Beuther, et al. A&A (2019)

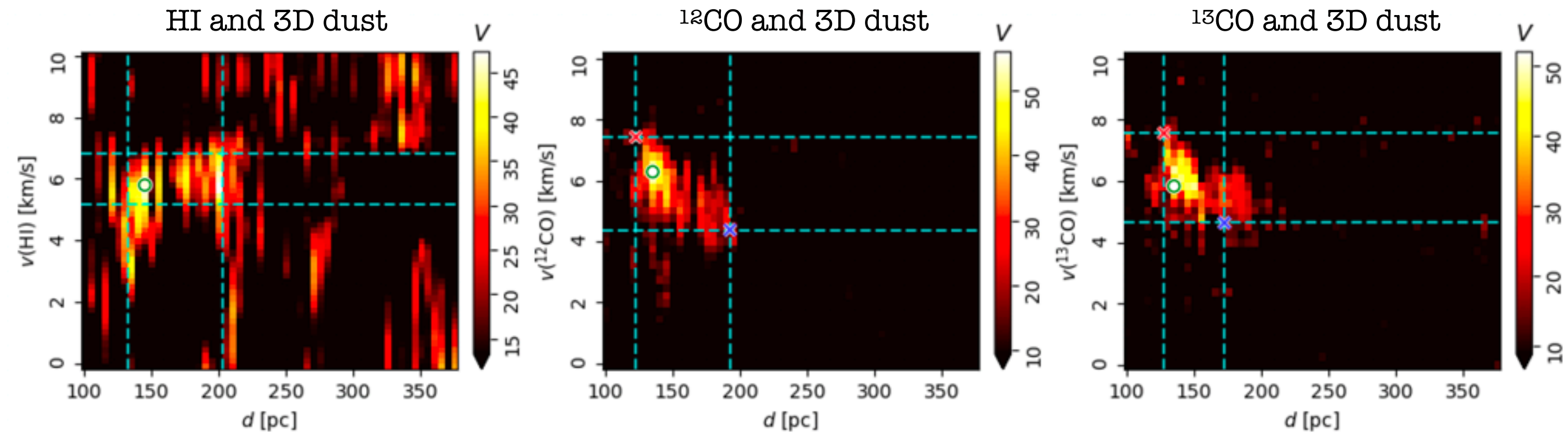


$$V = \frac{\sum_k^N w_k \cos 2\phi_k}{\sqrt{\sum_k^N (w_k)^2 / 2}}$$

Projected Rayleigh statistic
Jow et al. MNRAS (2018)

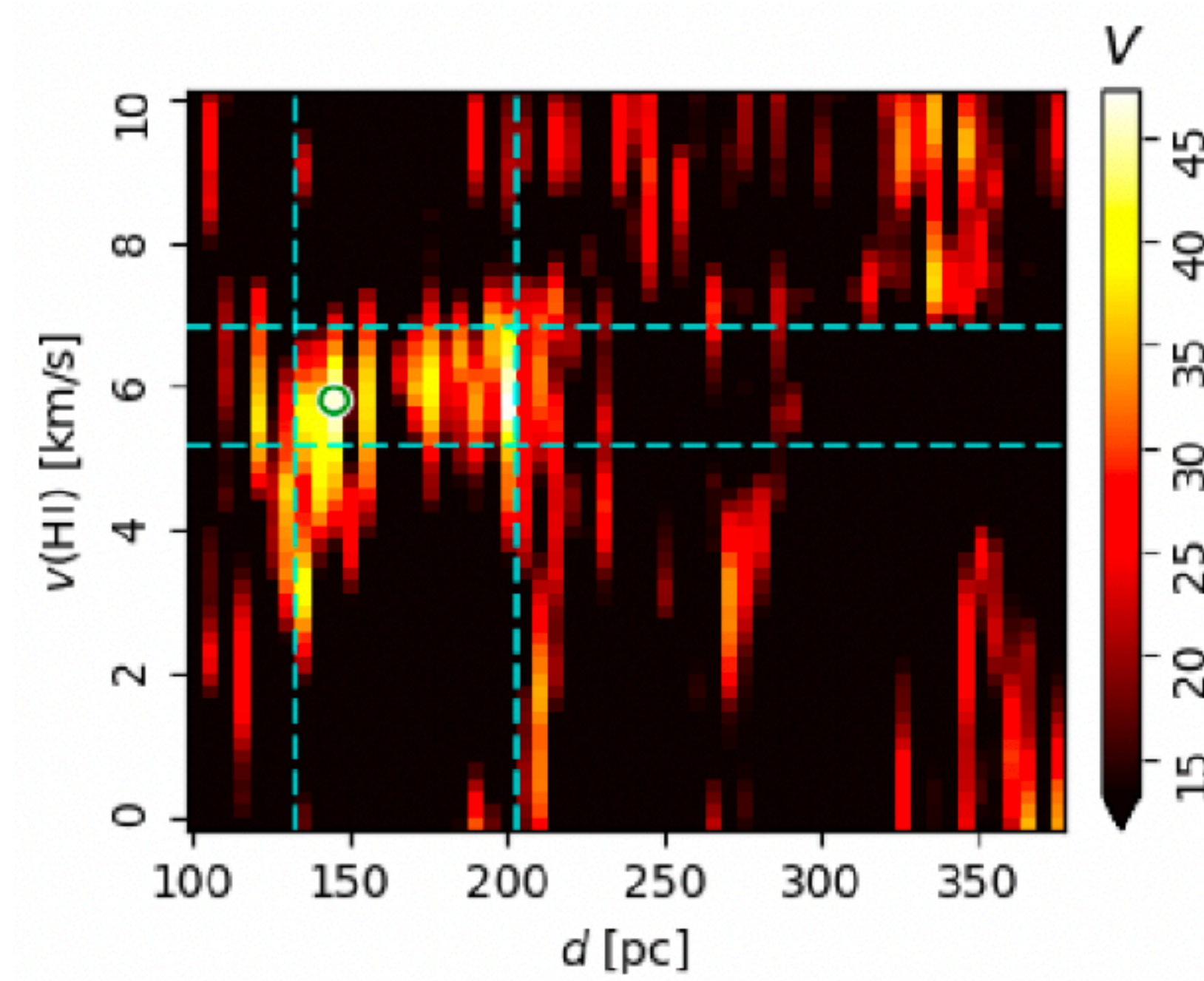
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Soler, Zucker, et al. A&A (2023)



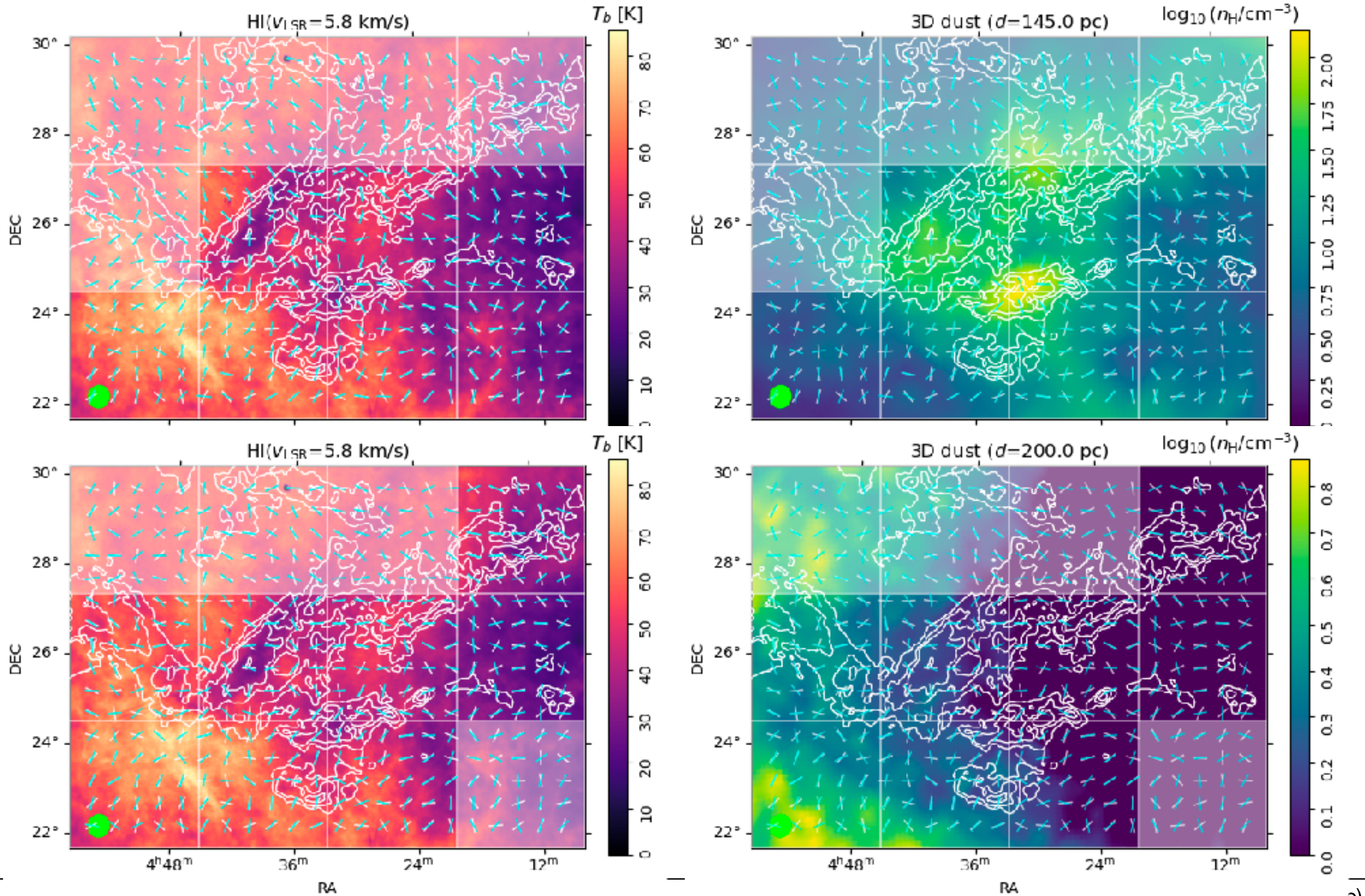
Taurus MC: HI and 3D dust

Soler, Zucker, et al. A&A (2023)



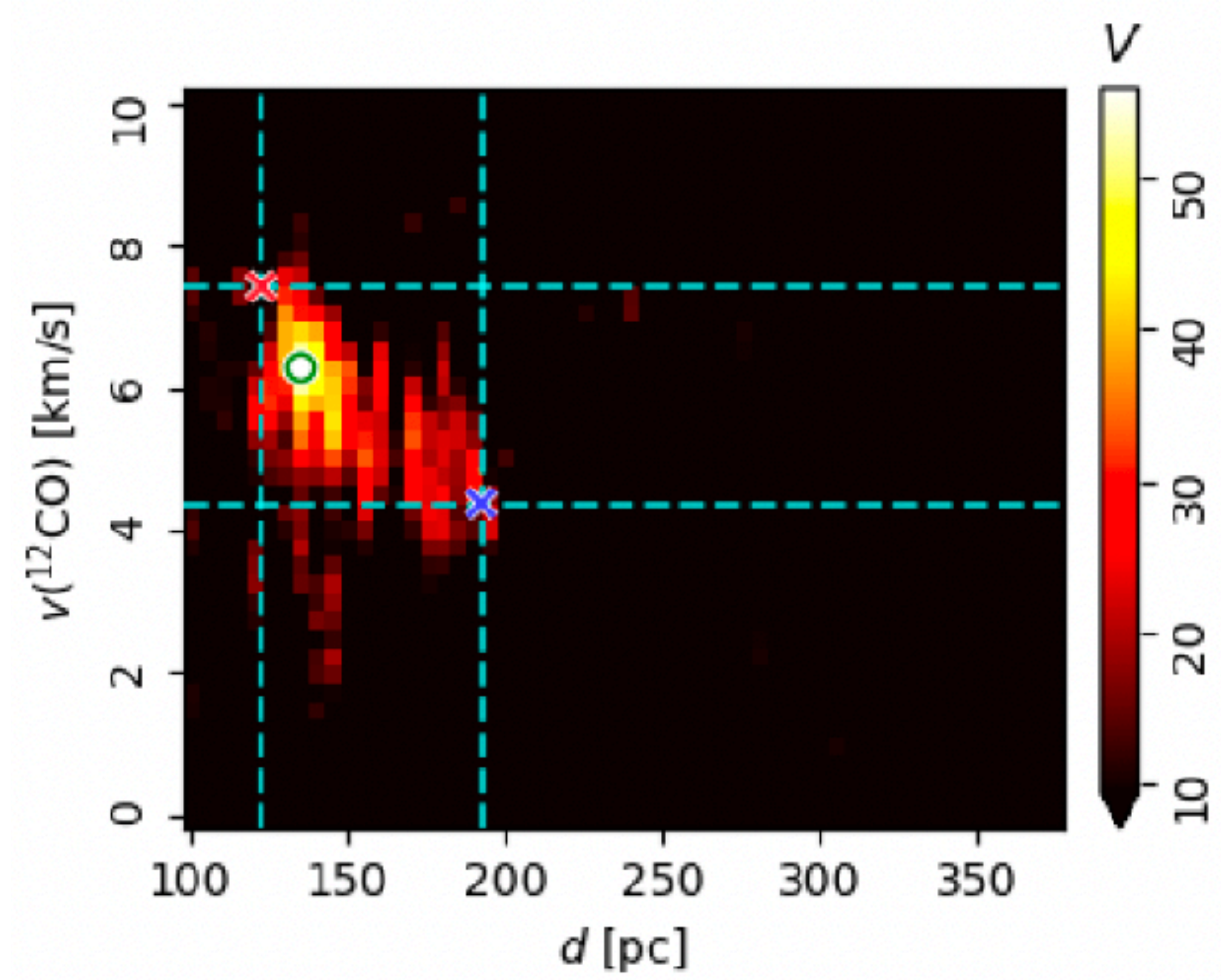
Taurus MC: HI and 3D dust

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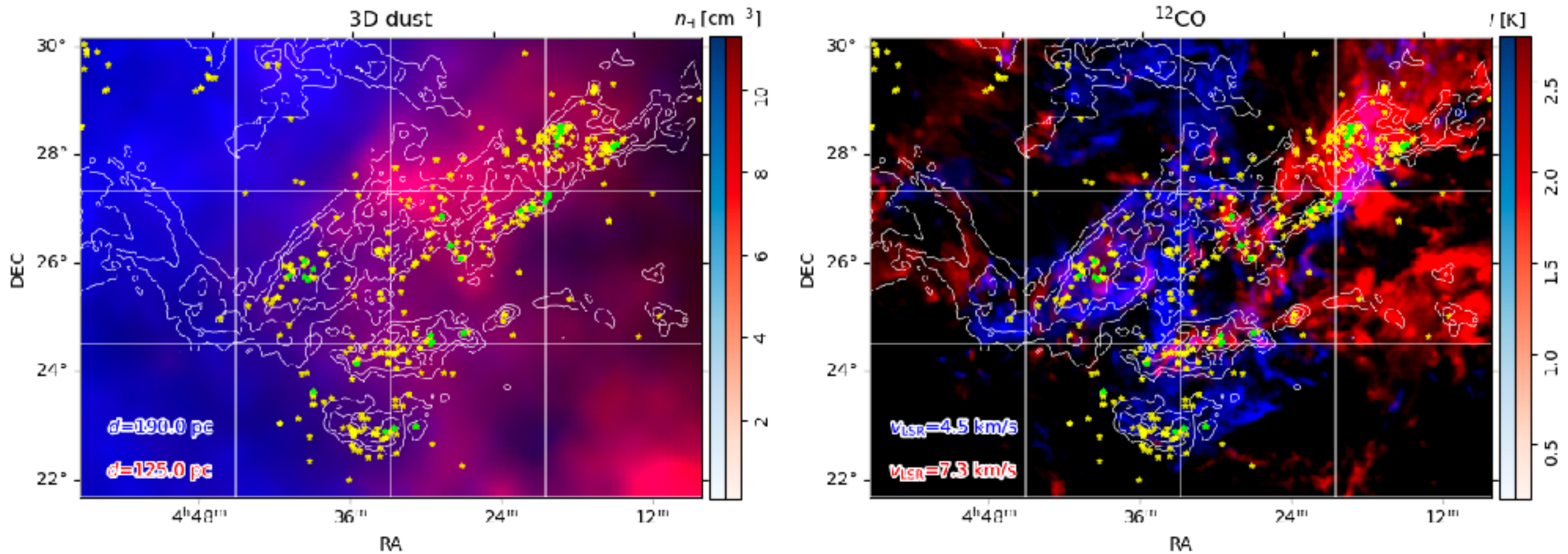
Taurus MC: ^{12}CO and 3D dust

Soler, Zucker, et al. A&A (2023)



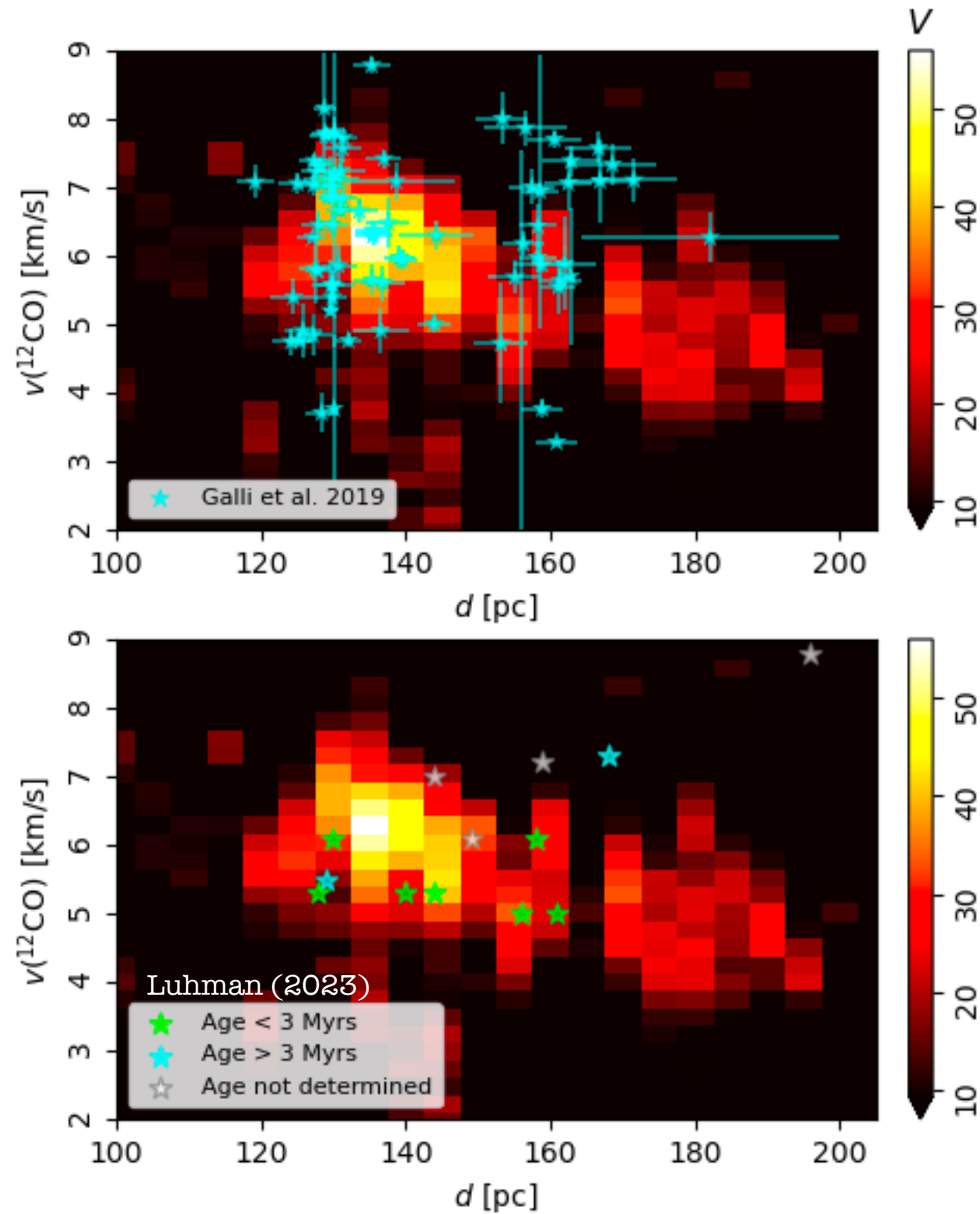
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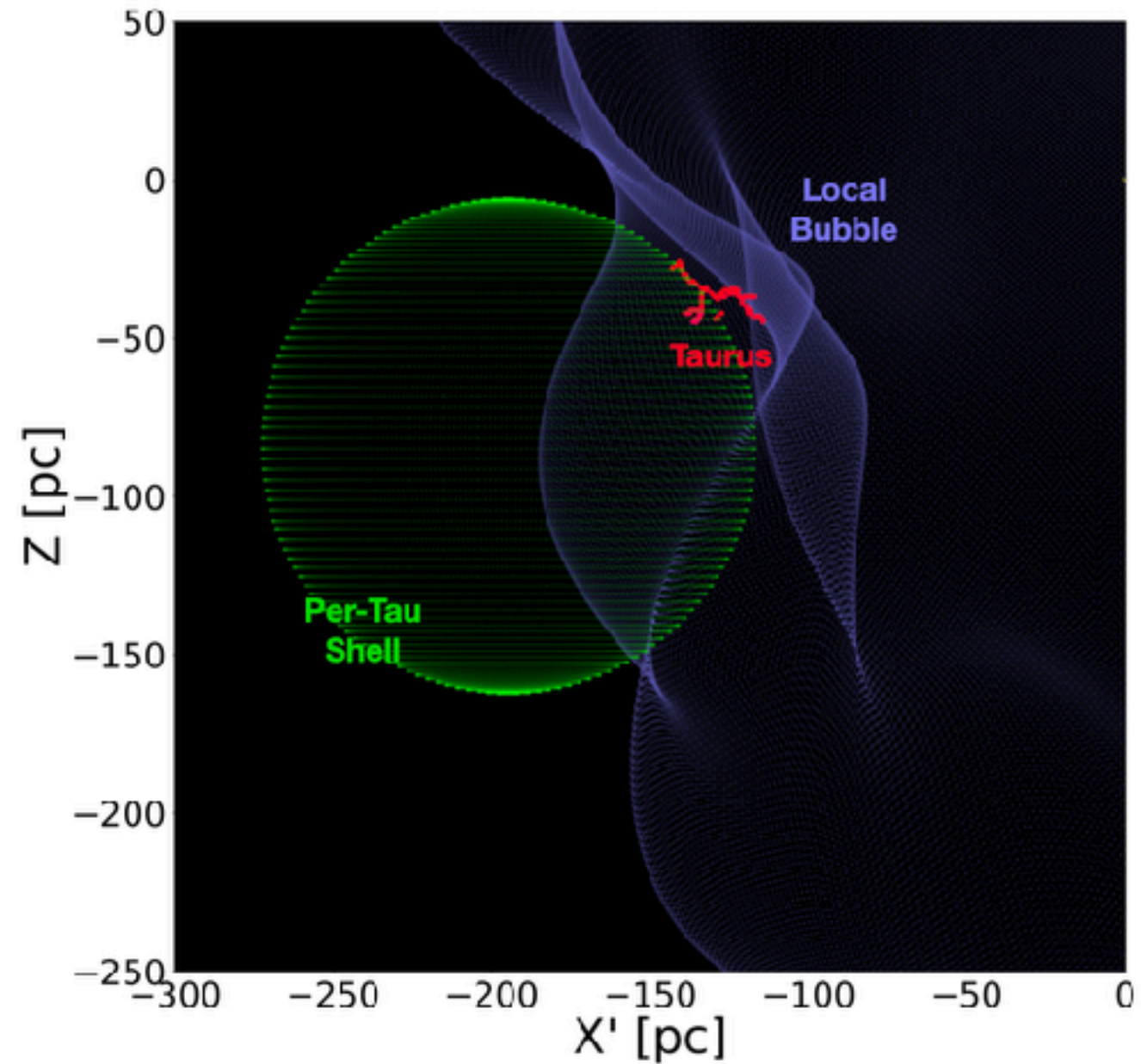
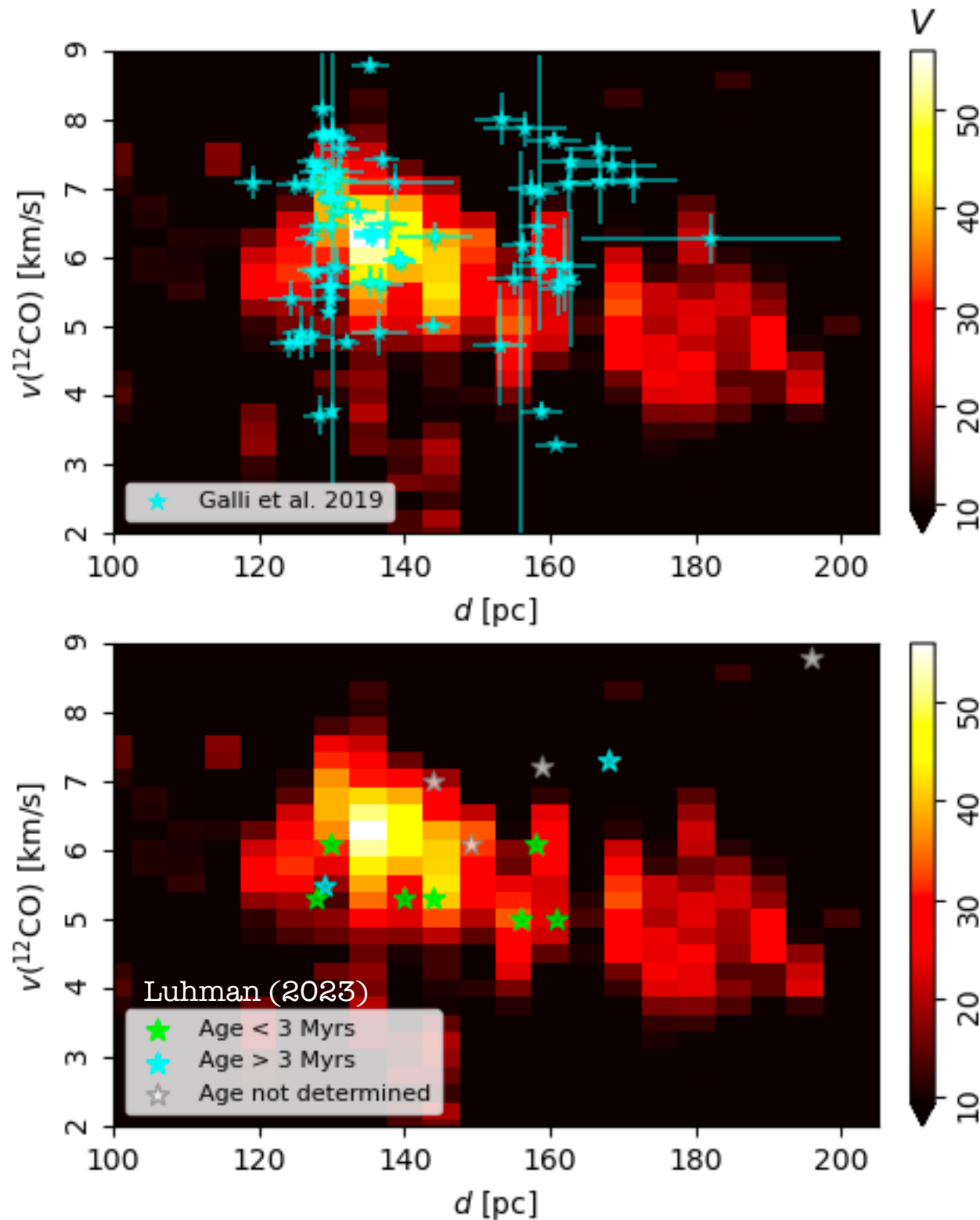
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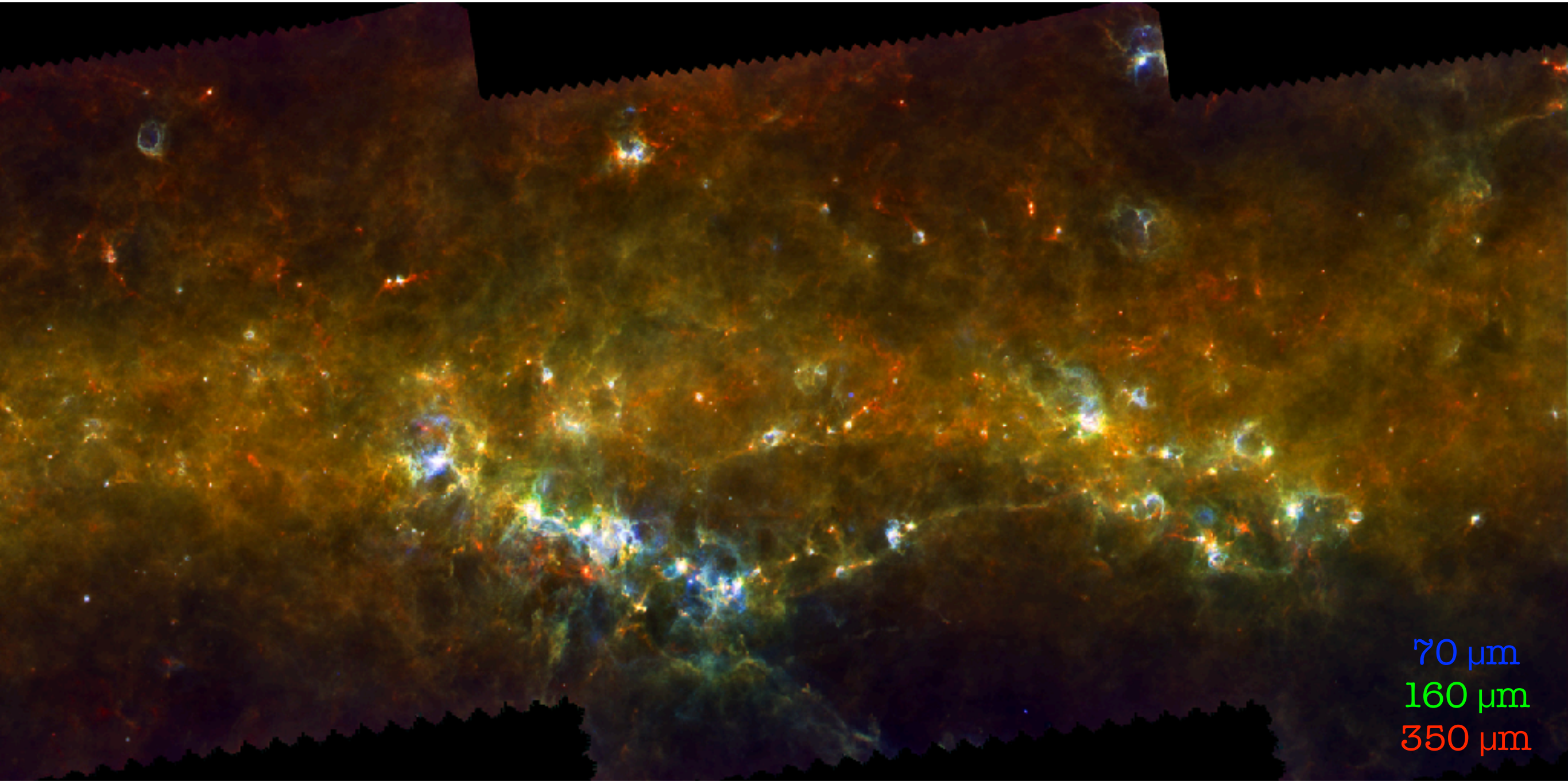
Taurus MC: ^{12}CO and 3D dust

Soler, Zucker, et al. A&A (2023)



Herschel infrared Galactic Plane Survey (Hi-GAL)

Molinari, Schisano, Elia et al. A&A (2014)

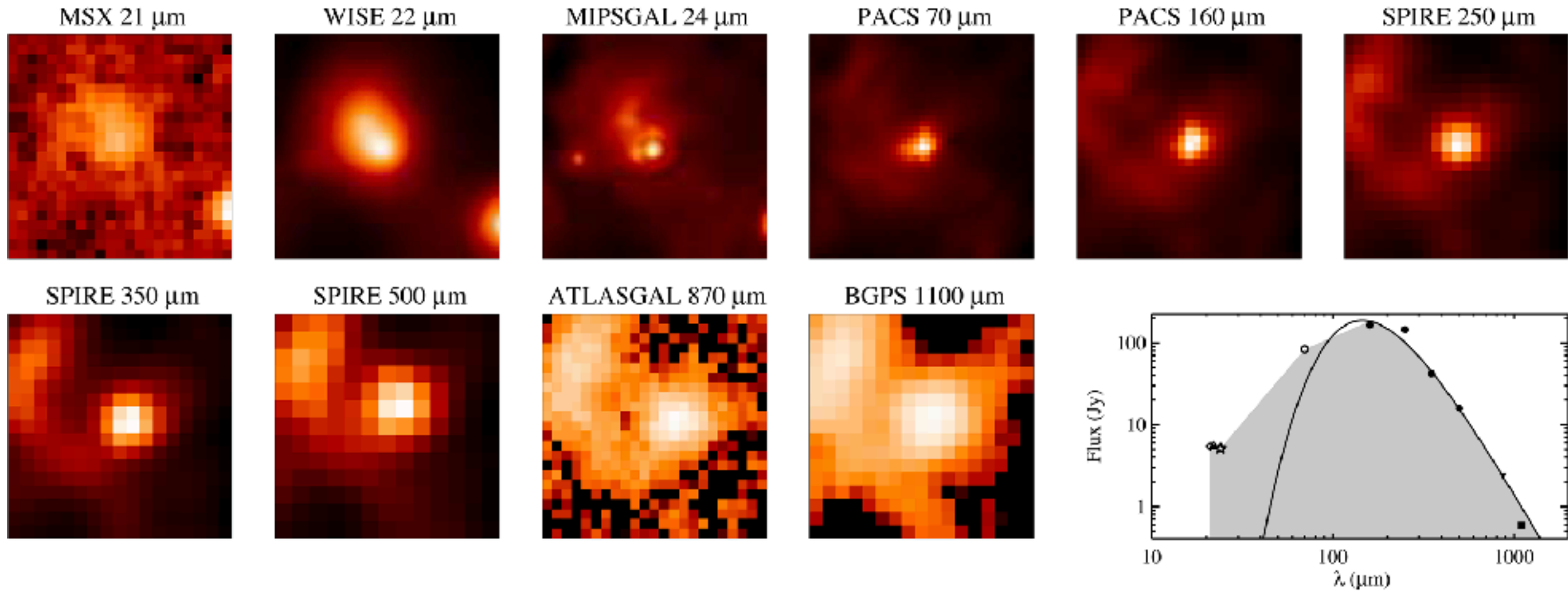


70 μm
160 μm
350 μm

Star formation from dust thermal emission

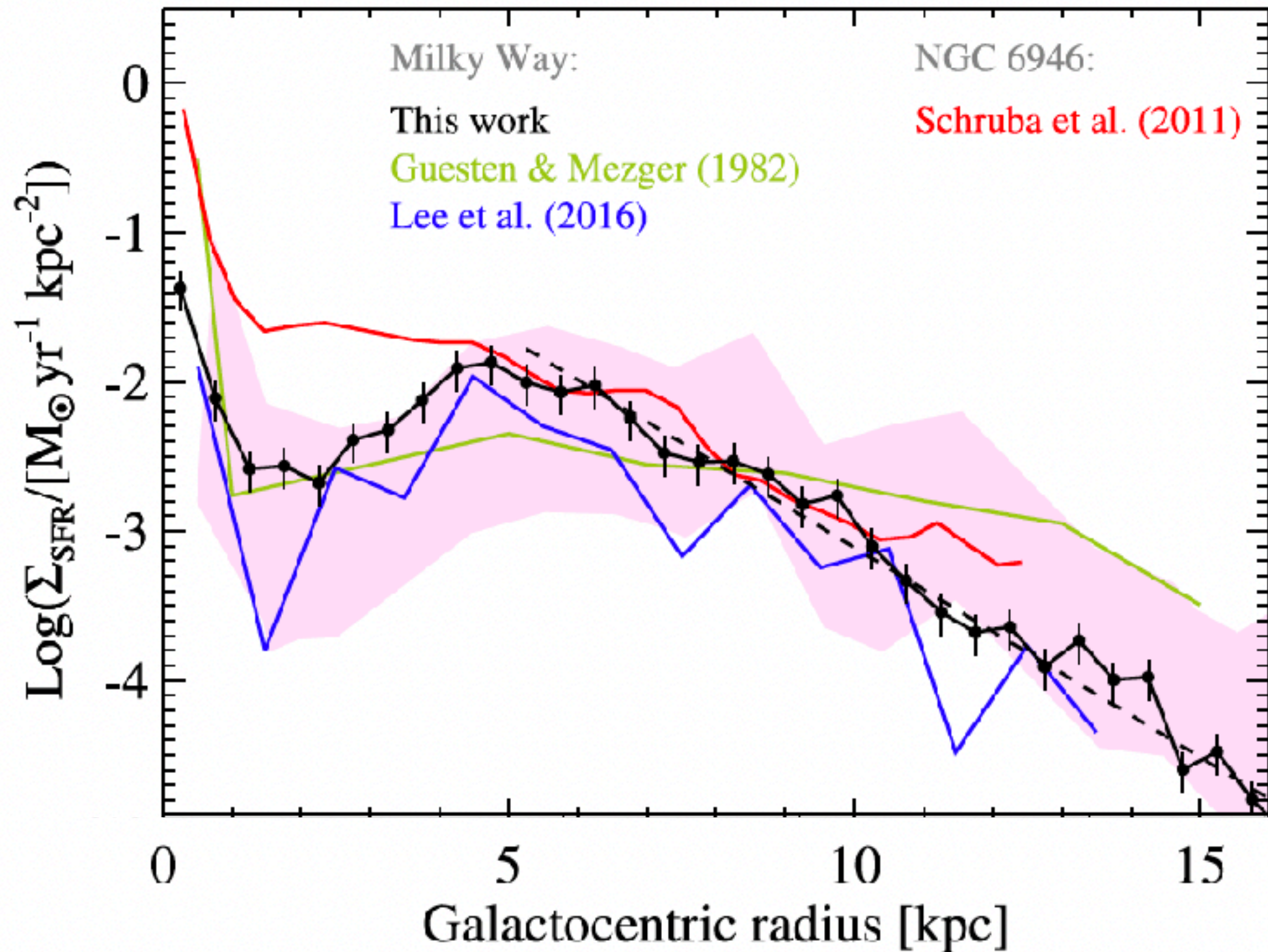
Elia, Molinari, Schisano, et al. MNRAS (2017)

Source #110522 (proto-stellar), $l = 24.7295^\circ$, $b = 0.1537^\circ$, $d = 9170$ pc, $M = 1317 M_\odot$, $T = 23.9$ K, $L = 21225 L_\odot$



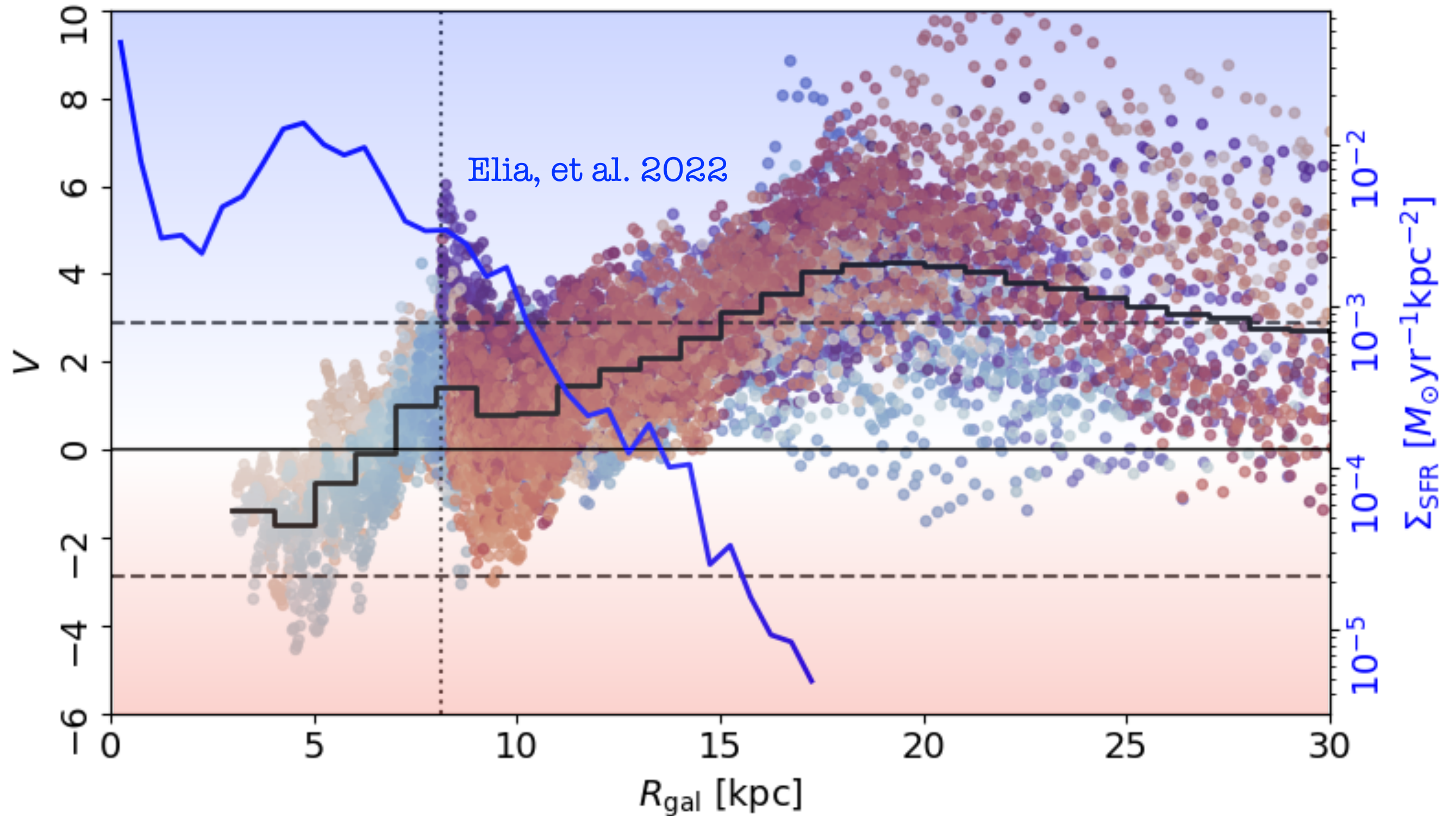
Star formation from dust thermal emission

Elia, Molinari, Schisano, Soler et al. A&A (2022)



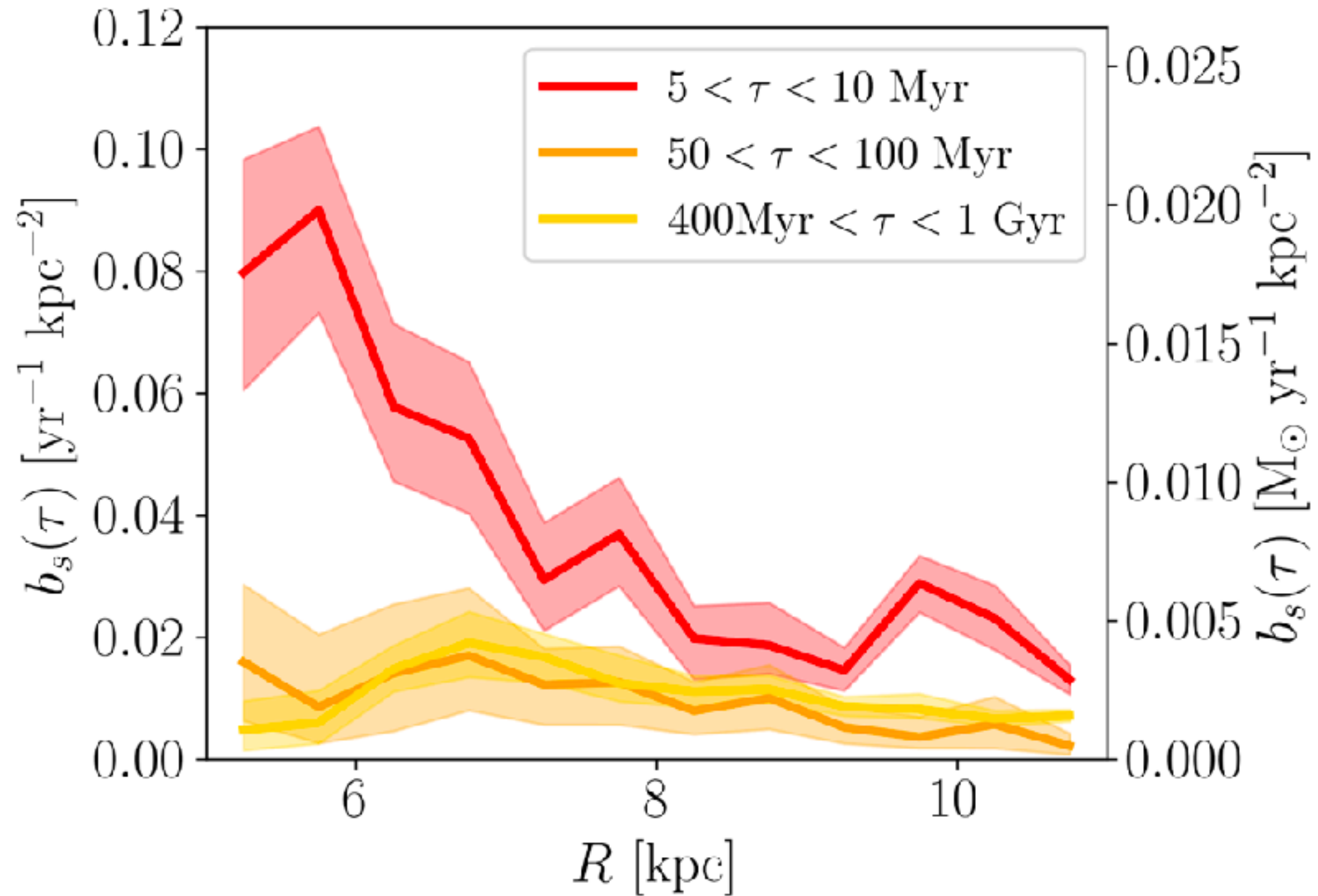
Atomic filament orientation and star formation

Soler, J.D. et al. A&A (2022)



SF from high-mass stars

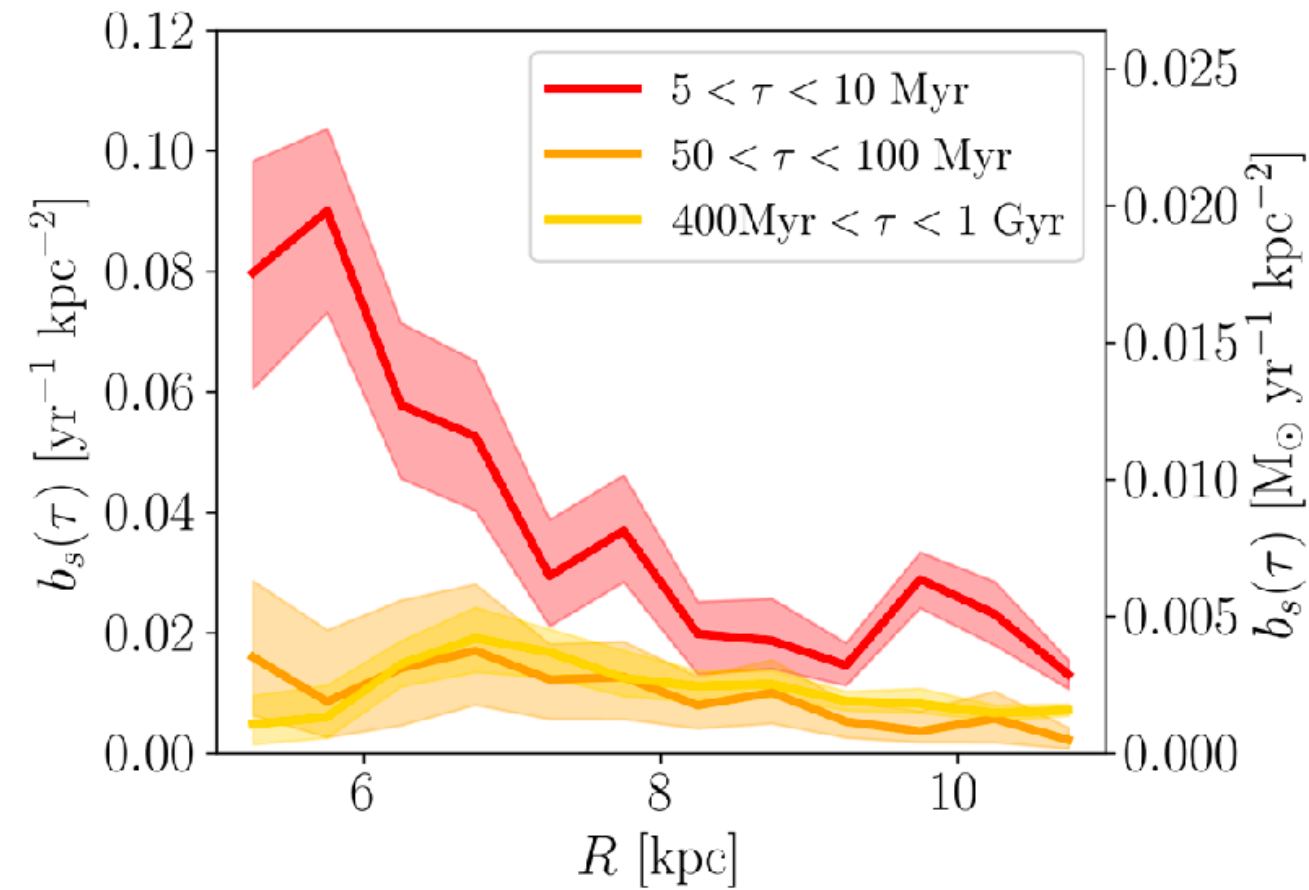
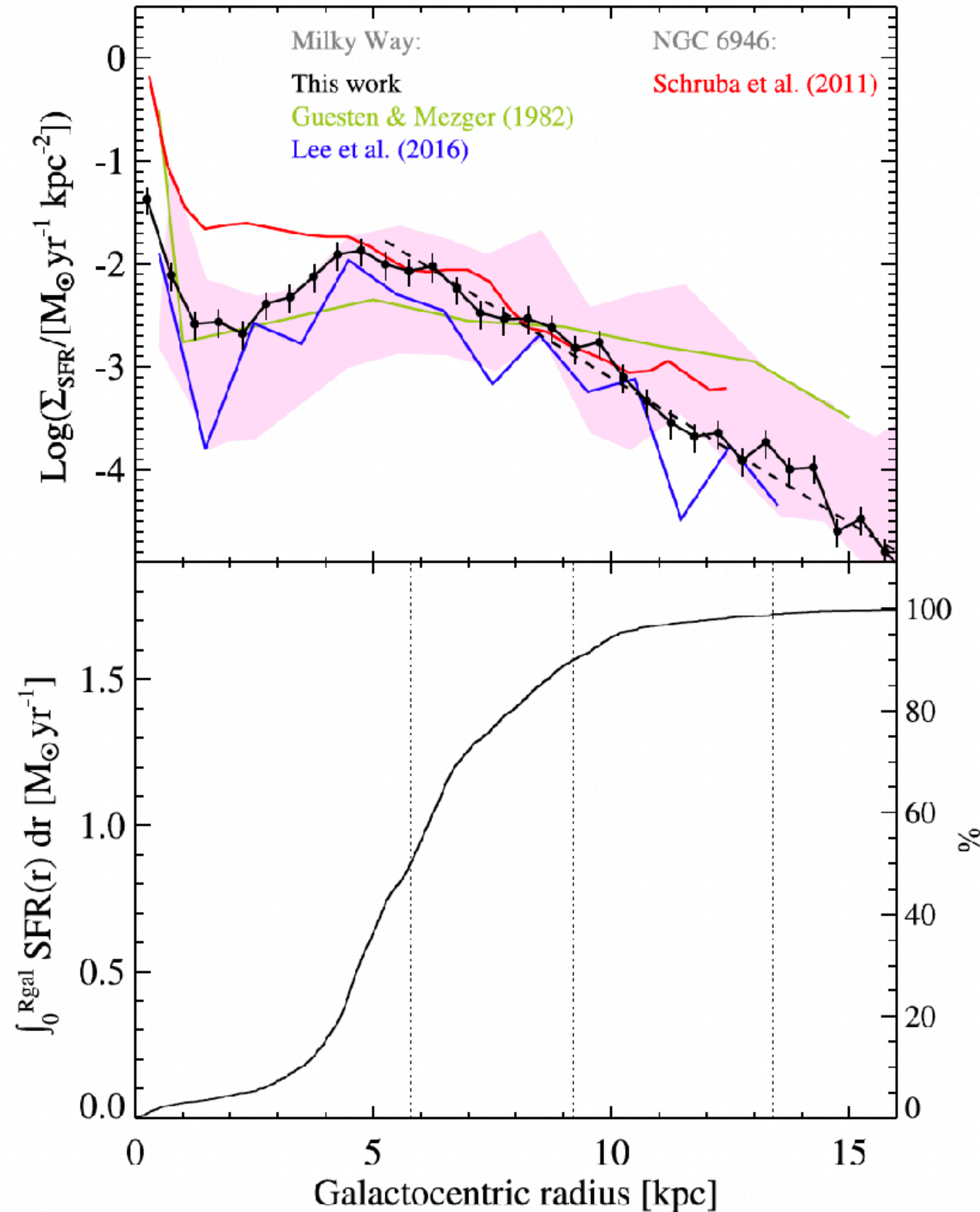
Zari, Frankel & Rix. *A&A* (2023)



SF from dust thermal emission and high-mass stars

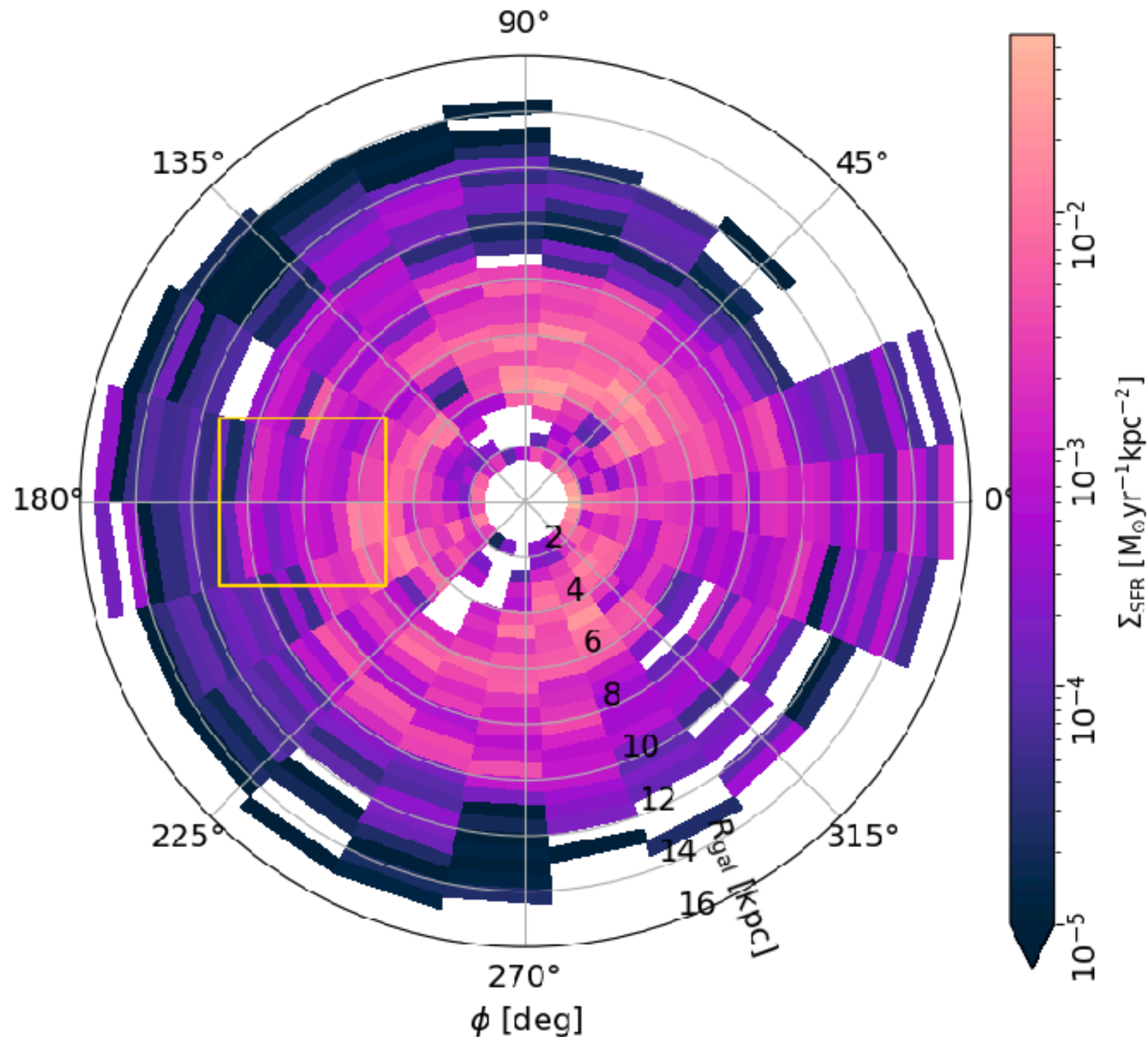
Elia, Molinari, Schisano, Soler et al. A&A (2022)

Zari, Frankel & Rix. A&A (2023)



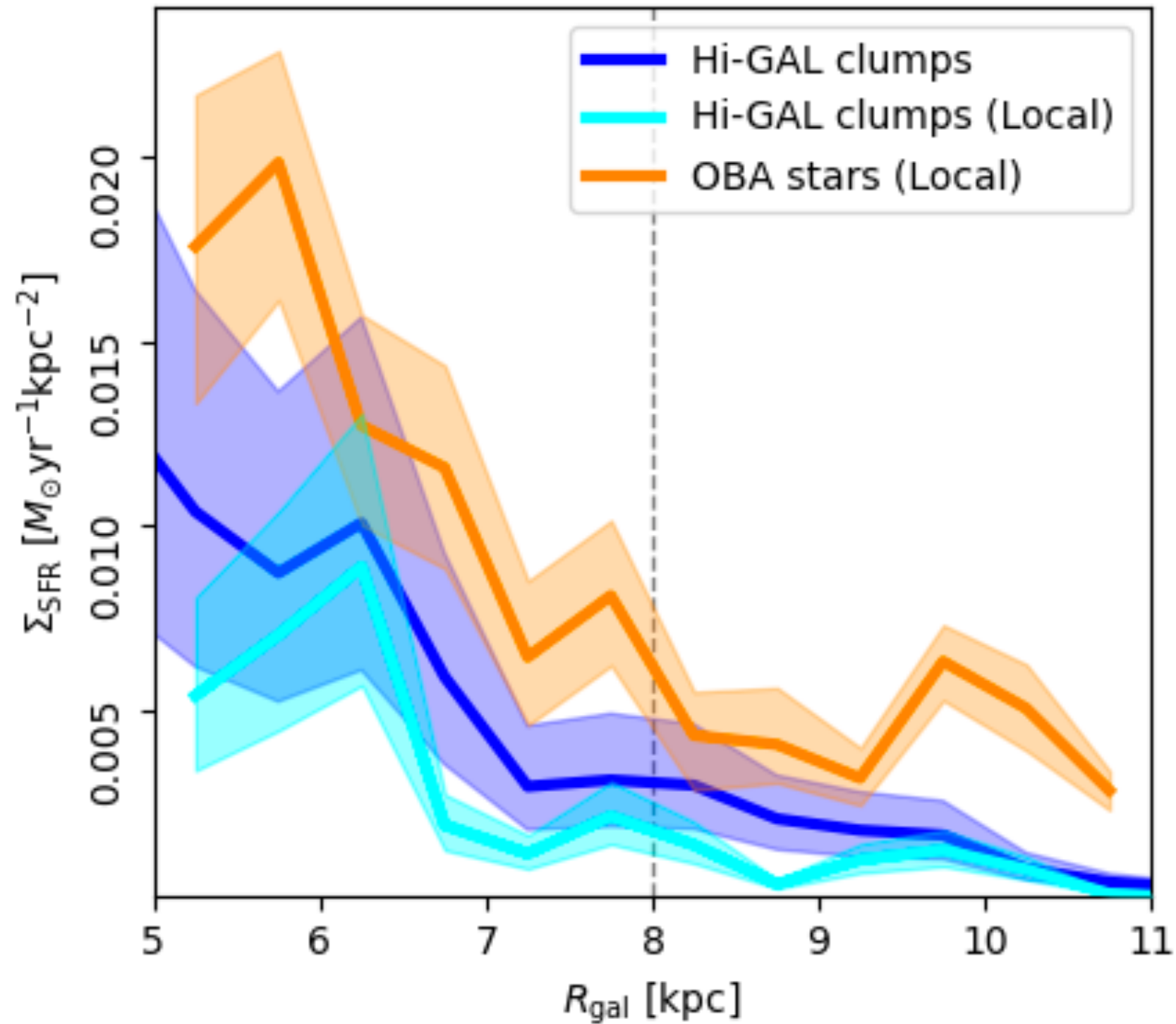
SF from dust thermal emission and high-mass stars

Soler, Zari, Elia et al. *A&A* submitted.



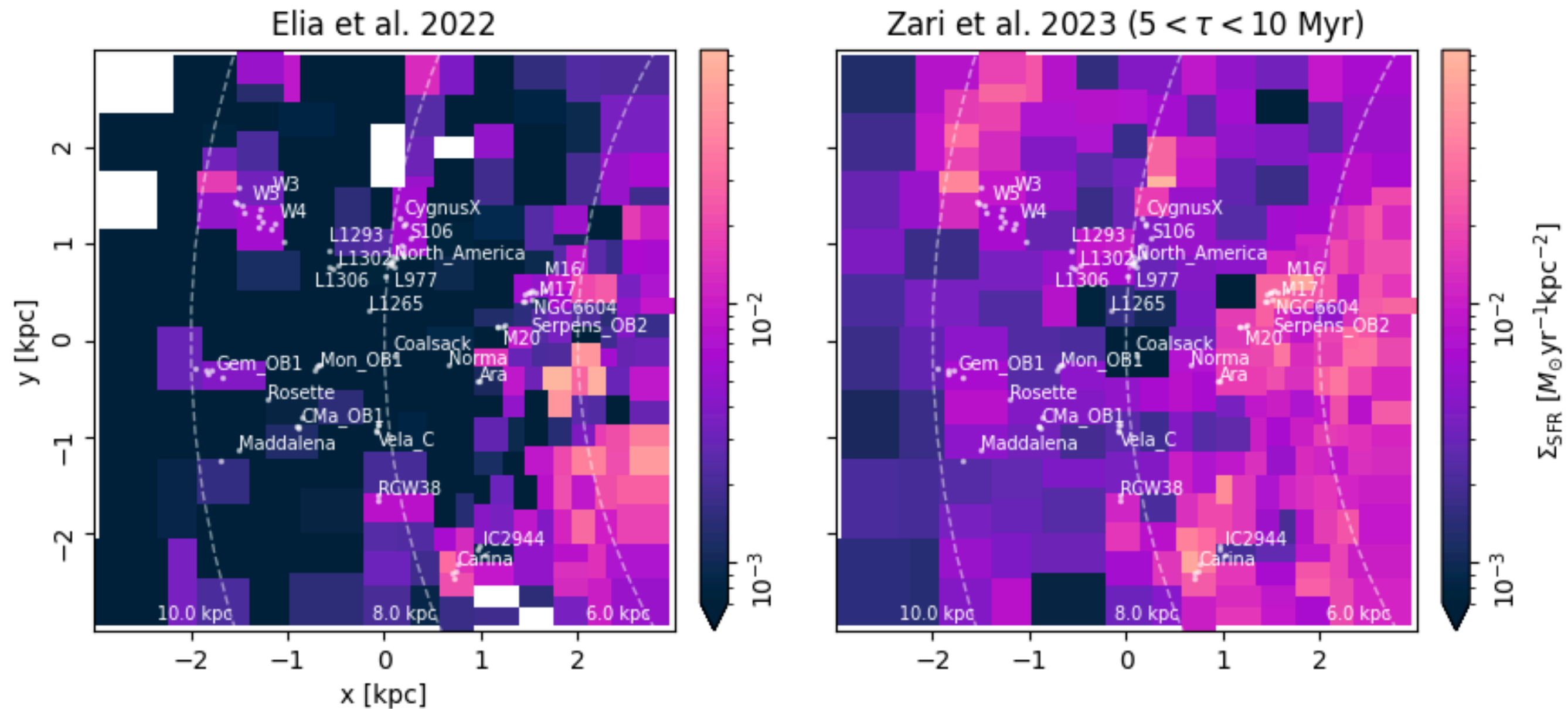
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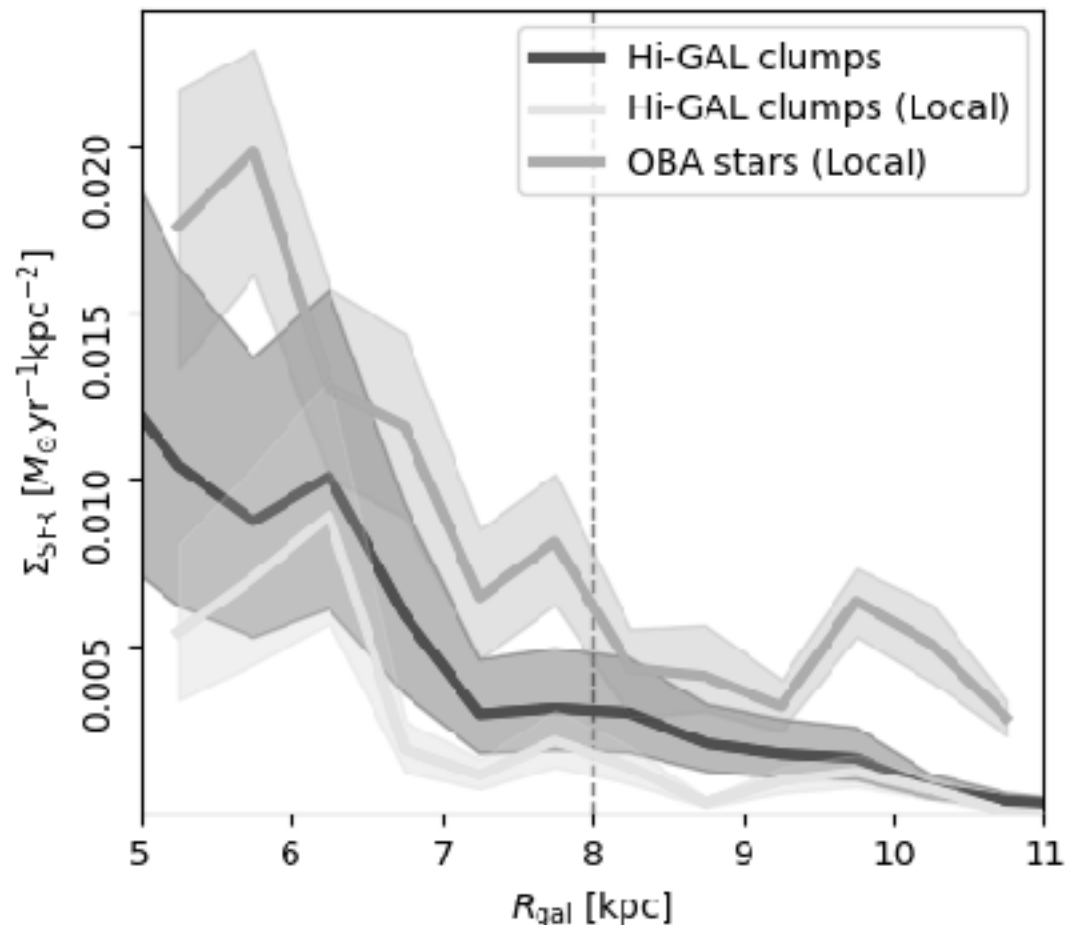
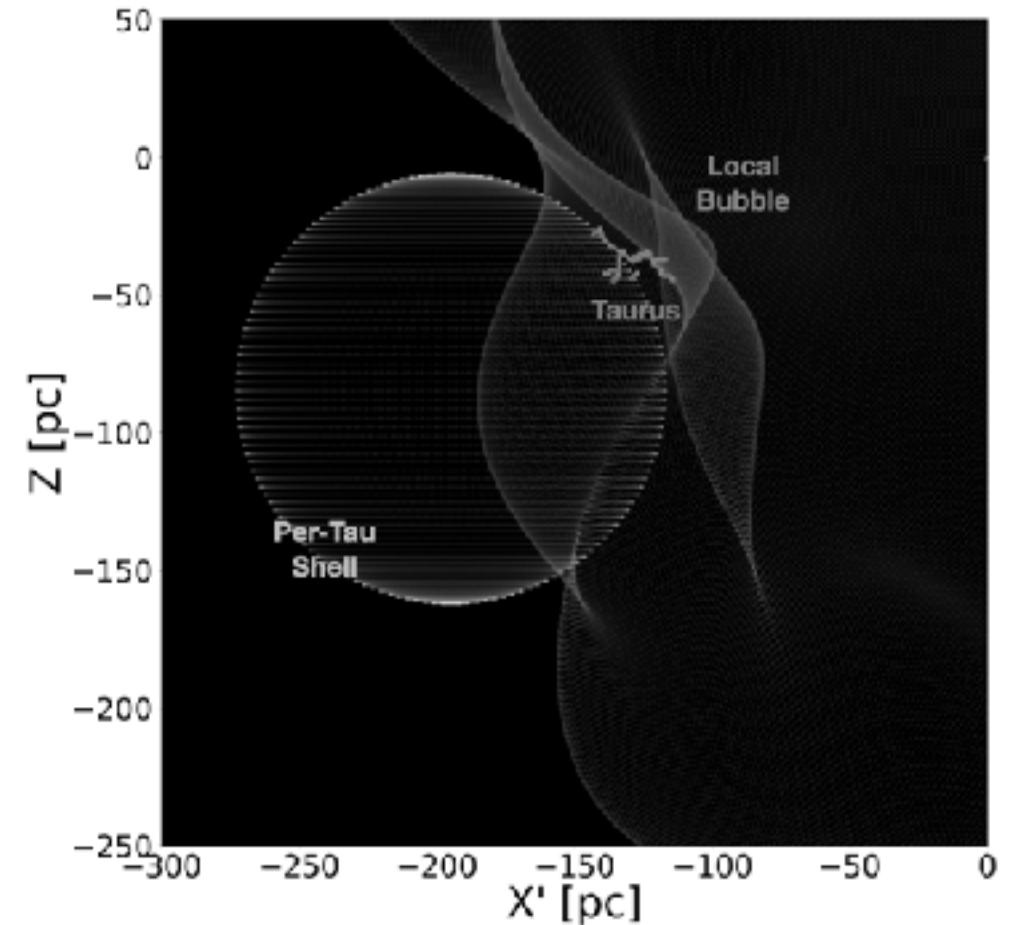
Conclusions

Dynamics of the Taurus molecular cloud

Soler, Zucker, Peek & the ECOGAL collaboration
A&A (2023)



The Taurus MC is experiencing compressive motions that are not followed by the YSOs in that region. [#StellarFeedback](#)



Star formation from dust thermal emission and high-mass stars

Soler, Zari, Elia et al. A&A submitted.

SF estimates from [#DustThermalEmission](#) and [#HighMassStars](#) agree (within a factor of 2) and indicate that MW's [#SFR](#) has been constant over last 10 Myrs.

